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BACHELOR THESIS

Physiological and geodata analyses of methane reduction potential in ruminants

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ABSTRACT

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LIST OF ABBREVIATIONS

6-Aza	6-Azaauracil.
8-Aza	8-Azahypoxanthin.
AA	aminoacids.
AB	antibiotic.
CFU	colony forming unit.
CRT	common reporting table.
Cyc-Se.	cycloserin.
DE	digestibility of food.
DEU	Germany.
DMSO	dimethylsulfoxid.
EnterFe	enteric fermentation.
<i>E. coli DH5α</i>	<i>Escherichia coli DH5α</i> .
<i>E. coli</i>	<i>Escherichia coli</i> .
EtOH	ethanol.
GE	gross energy intake.
LULUCF	land use and land use change and forestry.
<i>M. boviskoreani</i>	<i>Methanobrevibacter boviskoreani</i> AbM4.
<i>M.</i>	<i>Methanobrevibacter</i> .
<i>M. millerae</i>	<i>Methanobrevibacter millerae</i> ZA-10 ^T .
<i>M. thaueri</i>	<i>Methanobrevibacter thaueri</i> CW ^T .
<i>M. wolinii</i>	<i>Methanobrevibacter wolinii</i> SH ^T .
NZL	New Zealand.
OD600	optical density at 600 nm.

Puro	puromycin.
Sqr.	square.
USA	United States of America.
Ym	methane conversionfactor.

1 INTRODUCTION

The following section presents the results regarding a potential mitigation strategy against methane emissions: the immunization of cattle ([Baca-González et al., 2020](#)). As explained above, the bachelor's thesis - and thus the results - can be divided into two main sections:

1. The biochemical part, which involves a fundamental analysis of methane-producing archaea isolated from the rumen of cattle, as well as an attempt to develop genetic tools for them.
2. The geographical part, which, based on [Buendia et al. \(2019\)](#) findings, examines how much methane can be reduced and what role this can play in combating climate change.

In the enteric fermentation process of ruminants CH_4 are formed through methanogenic archaea. One discussed mitigation strategy to reduce methane resulting from cattle farming is to immunize the cattle against these specific archaea ([Malyugina et al., 2025](#)). For developing a specific immunization strategy, the isolated methanogenic archaea strains from cattle rumen has to be well understood. Although some strains have already been isolated from the rumen of cattle, they have hardly been studied to date ([Baca-González et al., 2020](#)). This bachelor's thesis aims to address this issue.

In doing so, the focus will primarily be on two strains: *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). Besides these two main strains, some experiments were also done on *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter millerae* ZA-10^T (*M. millerae*). These two methanogenic archaea were also isolated from the rumen of cattle. Goal of the investigation, that took place, was to detect several basic characteristics and try to implement first genetic engineering tools.

1.1 Biochemistry

Bin mit der Karte so noch unzufrieden, auch wenn sie skizzenhaft, glaube ich schnell zeigt was alles grob bei unseren Konstrukten drauf sein sollte. Bin mir auch noch unsicher ob ich so eine Art Figure hier in die Einleitung mit reinpacke, bei der Einführung unserer Konjugations/ Transformations technick oder erst im result part -> vmtl hier aber sinnvoller

1.2 Geography

DE has a non linear relation towards GE (see eq. (4):6).

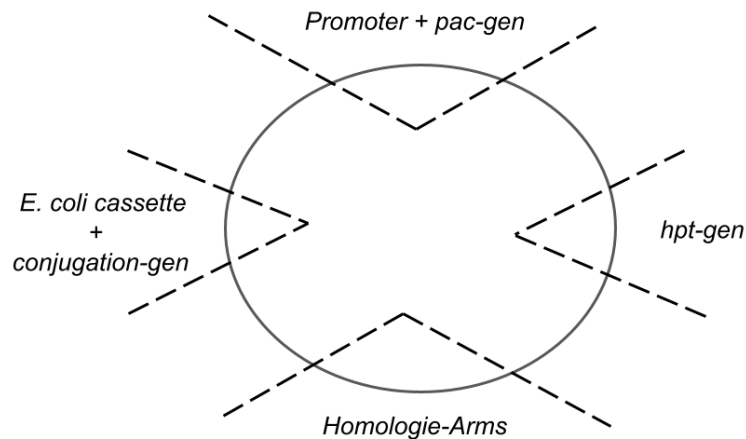


Figure 1.1: Sketch of the general structure of the suicide plasmids for the transformation of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) via conjugation using *E. coli*. The plasmid contains the *E. coli* cassette with *catp* as a selection marker using chloramphenicol, as well as the conjugation gene *traJ*. In addition, all plasmids contain the *hpt* gene to enhance counterselection and the two homology arms that flank the genomic *hpt* gene of the corresponding organism. The final important region is the one containing the *pac* gene and its associated promoter, which serves as a selection marker.

2 DATA AND MATERIALS

2.1 Biochemistry

2.2 Geography

2.2.1 Study area

The geodata analyses of methane reduction potential in rumineszenz will focus on the impact of the immunization of cattle. Two emission sectors of agriculture discussed in the [Buendia et al. \(2019\)](#) could be effected of this. On the one hand methane resulting from enteric fermentation and on the other hand from manure management. However, in this bachelor thesis the manure management sector will not be investigated, primarily due to as-yet-unclear links to a possible microbiome shift, as well as time constraints.

All calculations are based on the methodology described in [Buendia et al. \(2019\)](#) for calculating CH_4 emission from enteric fermentation in cattle.

Furthermore, is the calculation done for only three countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). These three countries were picked, because of

three main reasons.

First, the project partners for the biochemical research project to develop an immunization method of cattle against methanogen archaea are located in those very countries.

Second, all three countries belong of the ANNEX-I states. This means that, due to their larger economies and higher levels of consumption, they emit a significant amount of GHGs, including CH₄, and therefore bear a greater responsibility with regard to climate change and its mitigation.

Third, although all three are categorized as ANNEX-I states, there are some notable difference between this states, as e.g. in their size in terms of area, population, and economy (the US. and Germany as leading nations of the world) or in their predominant way to raise animals, particularly cattle (e.g. in New Zealand (NZL) pasteur grazing is dominant in Germany loose housing (?, p. 155; ?, Table 510)).

The time period considered in this bachelor's thesis is determined, on the one hand, by the data series from the published CRT (see section 2.2.2) and, on the other hand, by the evaluation of the mitigation strategy.

For the latter, the decade 2010–2019 is relevant, which is also relevant in [Saunois et al. \(2025\)](#), well as the year 2030 as the target year for the emission targets of the GMP.

2.2.2 CRT Data

Common reporting table (CRT) is the data source for the calculations performed. The CRTs were downloaded from the UNFCCC (?).

For Germany and New Zealand, the published data range from 1990 to 2024, and for the United States, from 1990 to 2022. Where is one CRT for one year is published.

For the analysis of methane emissions from enteric fermentation processes involving cattle, the “Table3” and “Table3.A” sheets of the .xlsx files were used. These show only the values for the year it was published for.

The “TableS3” sheet serves as the basis for analyzing methane emissions from the various sectors in each country as well as total emissions. This sheet lists not only the values for the current year but also all past years, so the CRT for 2024 and 2022 for the U.S. also includes all values for the remaining years.

In general, the published figures are always valid for or calculated based on the entire year.

Germany and New Zealand have defined two subcategories for cattle according to [Buendia et al. \(2019\)](#): “dairy cattle” and “non-dairy cattle.” The United States, on the other hand, chose to define its own subcategories for cattle, so that “dairy cattle” and “non-dairy cattle” were further subdivided. Section [3.2.1](#) explains how the data was processed to ensure that all three countries are comparable with one another.

The used variables from the CRT table are given in following units for each animal category:

- CH₄ emission in kt CH₄
- Population size in 1000s
- GE in MJ head⁻¹ year⁻¹
- Ym in %
- DE in %

2.3 Program

All analyses and processing of the data as well as the visualization took place in R (version 4.6.1). The main package used for processing and analyses was "tidyverse" and "purrr". The visualization was performed via "ggplot". For more information on that, check the R scripts.

3 METHODS

3.1 Biochemistry

3.2 Geography

The general methodology for the processing, analyses and visualization of the CRT data in R was to build a function and then apply these with the interested variables and data from the specific countries.

In the following part, the general procedure is described in the part below. For a more detailed overview, please check the R scripts.

3.2.1 Processing CRT-Data

Cattle enteric fermentation

For Germany and New Zealand all given data for cattle and its subcategories "dairy cattle" and "non-dairy cattle" were imported from the "Table3.A" sheet of the CRT tables for each year, and processed so, that for each year of each country a table was created with one column with the year, one column, with the cattle categories and the following columns are for the measured or calculated values. Afterwards, all one year tables of one country were merged, so that one country has one table, where the values for the cattle categories from 1990 to the last published year are present.

For the U.S. there are no direct "Dairy cattle" and "Non-dairy cattle" categories, but more detailed one. So after the import of the same value columns for each cattle category as described before, these more detailed subcategories have to be merged in the same 2 as for Germany and New Zealand. For this purpose, the official definition and classifications from the U.S. (??, Chapter 5, p. 6) were used. DE, GE and Ym were merged as a weighted mean with the populations as weight. The methane emission were summed.

Methane emission for each sector

For sector-specific emissions, only the "TableS3" sheet was used for the most recent CRT (for the U.S., the year 2022; for Germany and New Zealand, the year 2024). Here, all sectors and their subcategories were imported for all years, as were the total emissions. After processing the data, only the methane emission values for the sectors in general were retained, as well as the total methane emissions for each country.

3.2.2 Calculation of methane emissions from enteric fermentation processes in cattle

The total methane emission for cattle and its subcategories dairy and non-dairy was not directly extracted from the published common reporting table (CRT) for the countries, but calculated from the indicators methane conversionfactor (Y_m), gross energy intake (GE) and the population number of the animal categories. Therefore, the CH_4 emissions were calculated for dairy and non-dairy cattle following the eq. (1). The methane emission for cattle than is the sum of dairy and non-dairy.

$$CH_{4,T} = \frac{GE \cdot \left(\frac{Y_m}{100}\right) \cdot 365}{55.65} \cdot \frac{N_T}{10^6} \quad (1)$$

where:

- $CH_{4,T}$ = yearly methane emission of a animal category [kt CH_4 year⁻¹]
- GE = gross energy intake [MJ head⁻¹ day⁻¹]
- Y_m = methane conversion factor [%]
- 55.65 = energy content of CH_4 [MJ kg⁻¹]
- N_T = population-size of animal category
- 10^6 = conversion-factor kg to kt

3.2.3 Share of cattle methane emissions from enteric fermentation

The share of enteric fermentation was claculated for its fraction on the total amount of anthropogen produced CH_4 in a country and on the methane emissions resulting from the agriculture sector. For this purpose the published data in the CRT was used for each country.

3.2.4 Calculation of saved methane

To calculate the potential methane savings resulting from the mitigation strategy under investigation (vaccination of cattle), various success scenarios are defined: 10, 20, 30, 40, and 50

These will have an impact on two indicators used in [Buendia et al. \(2019\)](#). Y_m , which represents the proportion of energy or feed consumed by cattle that is converted into methane by that microbiome, and DE, assuming that the reduced amount of feed energy converted into methane now benefits the animal during digestion, meaning that the percentage of feed

digestibility for the animal increases.

The values of the resulting Ym (Ym_{Red}) and DE (DE_{Red}) are calculated as in eq. (2) and eq. (3).

$$Ym_{Red} = Ym \cdot \left(1 - \frac{Red}{100}\right) \quad (2)$$

where:

- Ym_{Red} = adjusted Ym for the investigated reduction scenario [%]
- Ym = original methane conversion factor from CRT [%]
- Red = investigated reduction scenario [%]

$$DE_{Red} = DE + \left(Ym \cdot \frac{Red}{100}\right) \quad (3)$$

where:

- DE_{Red} = adjusted DE for the investigated reduction scenario [%]
- DE = Digestibility of feed as a fraction of gross energy intake [%]
- Red = investigated reduction scenario [%]

The eq. (1) presents the relation of Ym and methane emission of cattle already. DE, on the other hand, is named in three different equations in the calculation of Buendia et al. (2019), shown in the eq. (4) through (6) with little adjustments. It has his impact on produced methane through his role in calculate the GE of the animal, which has a linear relation towards methane emission (see eq. (1)).

$$GE = \frac{\left(\frac{NE_x}{REM}\right) + \left(\frac{NE_y}{REG}\right)}{\left(\frac{DE}{100}\right)} \quad (4)$$

where:

- GE = gross energy intake [$\text{MJ head}^{-1} \text{ day}^{-1}$]
- NE_x = sum of the net energy required for maintenance (NE_m), for the activity (NE_a), for lactation (NE_l), for the work (NE_{work}) and for the pregnancy (NE_p) of the animal [$\text{MJ head}^{-1} \text{ day}^{-1}$]
- REM = ratio of net energy available in a diet for maintenance (Equation (5))

- NE_y = sum of the net energy required for growth and for wool production [MJ head⁻¹ day⁻¹]
- REG = ratio of net energy available for growth in a diet (Equation (6))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REM = \left[1.123 - (4.092 \cdot 10^{-3} \cdot DE) + (1.126 \cdot 10^{-5} \cdot DE^2) - \left(\frac{25.4}{DE} \right) \right] \quad (5)$$

where:

- REM = ratio of net energy available in a diet for maintenance
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REM = \left[1.164 - (5.16 \cdot 10^{-3} \cdot DE) + (1.308 \cdot 10^{-5} \cdot DE^2) - \left(\frac{37.4}{DE} \right) \right] \quad (6)$$

where:

- REG = ratio of net energy available for growth in a diet
- DE = Digestibility of feed as a fraction of gross energy intake [%]

To calculate the new methane emission for cattle corresponding to the reduction scenario now, GE has to be recalculated with the adjusted DE. Afterwards eq. (1) can be used again to calculate the adjusted CH₄ emission with the adjusted GE (GE_{Red}) and Ym_{Red}.

For calculation the GE_{Red} DE_{Red} can not just simply be put in the eq. (4), because of the values for NE_x and NE_y. Although the values for the individual net energy intake categories (see eq. (4)) should not change, these are not given in the published CRTs of the countries.

To solve this problem, I assume two things:

1. The net energy required for wool production an irrelevant categorie is for the investigation of cattle.
2. Most of the animal in the population of dairy and non-dairy cattle are fully grown or close to it. Here, too, the IPCC only classifies calves "dairy" or "non-dairy" once they started eating solid food. Because of that the net energy needed for growth will overlooked and set to 0.

Now, we can slim down the eq. (4) and rearrange it to calculate a fixed value for NE_x for a specific year, because the part with NE_y turns zero (see eq. (7)).

$$NE_x = GE \cdot REM \cdot \left(\frac{DE}{100} \right) \quad (7)$$

$$GE = \frac{\left(\frac{NE_x}{REM} \right)}{\left(\frac{DE}{100} \right)} \quad (8)$$

where:

- NE_x = sum of the net energy required for maintenance (NE_m), for the activity (NE_a), for lactation (NE_l), for the work (NE_{work}) and for the pregnancy (NE_p) of the animal [MJ head⁻¹ day⁻¹]
- GE = gross energy intake [MJ head⁻¹ day⁻¹] (eq. (4))
- REM = ratio of net energy available in a diet for maintenance (Equation (5))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

Now, GE_{Red} can be calculated for a given year using the fixed values for NE_x and DE_{Red} , when NE_y is assumed to be 0 (see eq. (8)). The calculated gross energy intake is then substituted into eq. (1) along with the corresponding Ym_{Red} value. This yields the methane emissions ($CH_{4,R}$) for a cattle category under a reduction scenario for a specific year.

The saved methane emissions ($CH_{4,S}$) for a specific reduction scenario are calculated by subtracting the original value for methane emissions from the value adjusted for the scenario (see eq. (9))

$$CH_{4,S} = CH_{4,T} - CH_{4,R} \quad (9)$$

where:

- $CH_{4,S}$ = saved CH₄ through reduction scenario [kt CH₄ year⁻¹]
- $CH_{4,T}$ = yearly methane emission of a animal category [kt CH₄ year⁻¹]
- $CH_{4,R}$ = yearly methane emission of a animal category for a corresponding reduction scenario [kt CH₄ year⁻¹]

3.2.5 Prediction of CH₄ emission until 2030

Two steps are required to calculate total methane emissions, as well as methane emissions for three reduction scenarios (10, 30, and 50) through the year 2030.

First, total methane emissions without emissions from EnterFe processes in cattle are calculated and then extrapolated.

Second, to calculate the methane emission from enteric fermentation (EnterFe) in cattle, the values for DE, Ym, Population size and NE_x were predicted for each subcategory ("dairy" and "non-dairy cattle") until 2030. For that, NE_x had to be first calculated (see eq. (7)). From the predicted values for these indicators the cattle EnterFe CH_4 emission could be calculated (via eq. (8) and (1)).

Also the methane emission values for three different reduction scenarios (10, 30 and 50) were calculated with these values using the eq. (2), (3), (8) and (1).

Afterwards, the emissions of the subcategories were summed to the CH_4 emission for cattle resulting from enteric fermentation processes.

Third, the predicted emission values for total without cattle enteric fermentation were added with the calculated emission values for cattle enteric fermentation to achieve the total methane emission of a country.

The forecast was made using a linear regression, with the value to be predicted depending on the respective year. However, the regression was not based on the entire time period, but rather on three different time intervals (the last 5, 10, and 15 years). For each linear model, both the confidence interval (CI) and the total emissions were calculated. In addition, the R^2 value was determined for each model.

The extrapolation extends from 2025 to 2030 for New Zealand and Germany, and from 2023 to 2030 for USA.

4 RESULTS

4.1 Biochemistry

All following experiments were done with *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) and *Methanobrevibacter thaueri* CW^T (*M. thaueri*), in some *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) were investigated as well.

4.1.1 Cultivation

Media

For cultivation two liquid, complex media were tested, the DSM 1523 and the DSM 119. Both media are suitable for the cultivation of *M. thaueri* and *M. boviskoreani*, although some specific behavior could be detected.

The *M. thaueri* grew more slowly and not as dense as the *M. boviskoreani*. Also the *M. thaueri* started to lyse rapidly in a few days after full growth. This was observed much more prominently in DSM 1523 than in 119.

M. boviskoreani on the other hand did not show the described lysing behavior but precipitated in both media. These precipitates were not completely dissolvable through shaking. *Here I also could implement the Microscopy picture of the precipitate?*

Substrate

Besides the general cultivation with a H₂/CO₂ headspace, it was tested if *M. thaueri* and *M. boviskoreani* could use ethanol (EtOH) as an alternative source of reducing potential for growth, as a liquid option to hydrogen. For the tested concentration of EtOH (10 mM, 20 mM, 50 mM, 100 mM, 200 mM) no visual growth could be detected. *In einer literatur steht für typstamm bovis er wachse auf EtOH, und für Format auch -> das könnte man reintheoretisch noch testen*

4.1.2 Cell count

Could also show a microscopy picture of cell counting? -> hänge hier vmtl am besten noch eins in den Anhang, oder soll ich für alle 4 dann ein Bild in den anhang hängen?

Cell counting was done with a phase-contrast microscope and a Neubauer improved counting chamber, only the *M. wolinii* with an optical density at 600 nm (OD₆₀₀) of 0.387 was counted

with a Buerker counting chamber. In the Buerker, squares were counted, in the Neubauer improved, the group squares were counted, both have a volume of $4 \cdot 10^{-6}$ ml, as can be seen in LO - Laboroptik GmbH (2026). **der part hiervor vmtl eher in die methoden**

The cell-number per ml for the cultures was then determined using the following calculation:

$$\text{cellnumber} \cdot \text{ml}^{-1} = \frac{\varnothing \text{ cellnumber} \cdot \text{dilution factor}}{\text{Volume of the counted Square}} \quad (10)$$

Table B1 through B6 in the appendix contain the original counted cellnumbers per filled chamber and square. To calculate the cell count per ml, the arithmetic mean was determined for each culture (tab. B7) and substituted into the equation 10 along with the corresponding dilution factor (see ibid.). The results of the calculation were recorded in a table 4.1. The calculation was performed as an example using *M. wolinii* (OD600 = 0.387):

$$\text{cellnumber} \cdot \text{ml}^{-1} = \frac{216.375 \text{ cells} \cdot 20}{4 \cdot 10^{-6} \text{ ml}} = 1,081,875,000 = 10.819 \cdot 10^8$$

Table 4.1: Cellnumber per ml of different *Methanobrevibacter* (*M.*) culture for corresponding optical density at 600 nm (OD600). For counting the cells a phase-contrast microscope was used and a Buerker or Neubauer improved counting chamber at a 400x magnification. The cell count per ml was calculated using the equation 10, and the values were taken from the table B7.

culture	OD600	cellnumber · ml ⁻¹
<i>M. wolinii</i>	0.387	$10.819 \cdot 10^8$
<i>M. wolinii</i>	0.396	$10.672 \cdot 10^8$
<i>M. thaueri</i>	0.235	$5.603 \cdot 10^8$
<i>M. thaueri</i>	0.116	$3.644 \cdot 10^8$
<i>M. millerae</i>	0.378	$8.880 \cdot 10^8$
<i>M. boviskoreani</i>	0.287	$9.272 \cdot 10^8$

A clear trend can be observed for the cellnumber per ml of the different *Methanobrevibacter* (*M.*) culture throughout the table 4.1: higher OD600 values correspond to a higher cell count per ml. At the lowest measured OD600 of 0.116 (*M. thaueri*), $3.644 \cdot 10^8$ cells per ml was determined, and at the highest OD600 of 0.396 (*M. wolinii*), $10.672 \cdot 10^8$ cells per ml was determined.

Nevertheless, differences between the organisms are also apparent. For example, *M. boviskoreani* (OD600 = 0.287) has a higher cell count per mL ($9.272 \cdot 10^8$) than *M. millerae* (OD600 = 0.378 and cellnumber per ml = $8.880 \cdot 10^8$) despite having a lower OD600 of nearly 0.1. Furthermore, both *M. wolinii* cultures and *M. millerae* have similar OD600 values between 0.378 and 0.396, but those of *M. wolinii* are around $2 \cdot 10^8$ cells per ml higher (10.819 or $10.672 \cdot 10^8$ to $8.880 \cdot 10^8$ cells per ml).

The *M. wolinii* cultures are quite close to each other in terms of cell count per ml, with the higher value being determined for the culture with the lower OD600.

Besides *M. wolinii*, two cultures of *M. thaueri* were also examined to determine cell count. Here, one culture has an OD600 of 0.235 that is slightly more than twice as high as the other one with 0.116, but a cell count per ml that is slightly less than twice as high ($5.603 \cdot 10^8$ to $3.644 \cdot 10^8$ cells per ml).

4.1.3 Plating

Spread plating

100 μ l (nochmal im Laborbuch nachschauen) of a full-grown culture were spread plated as duplicates on a DSM 1523 solid media (1.5% agar) under anaerobic condition in a glovebox. In figure 4.1 the plating results are shown.

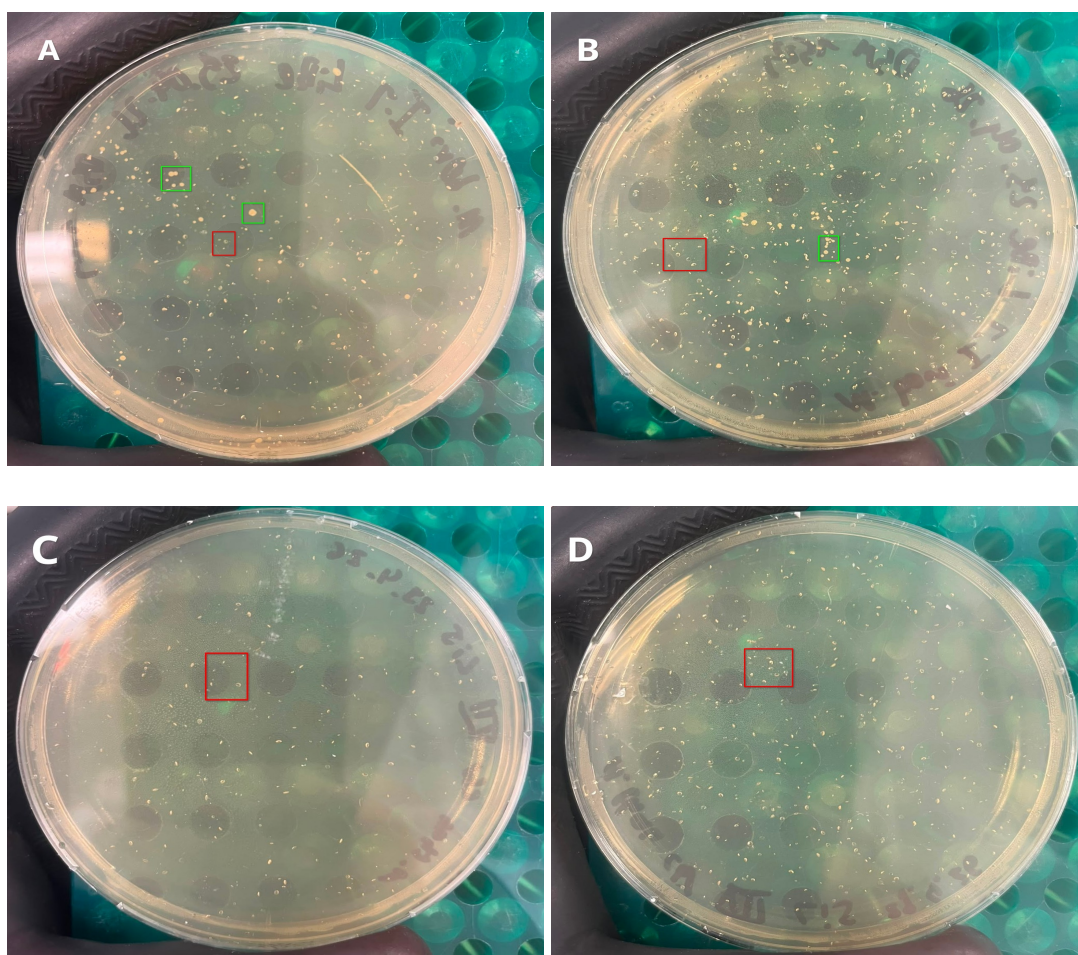


Figure 4.1: Spread plating results of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). 100 μ l of a full-grown culture was plated on DSM 1523 (1.5% agar) in duplicates. Red rectangles highlight gas bubble spots, green rectangles grown colonies. A: *M. boviskoreani*, B: *M. boviskoreani*, C: *M. thaueri*, D: *M. thaueri*

Various spots can be seen on all 4 plates. Most of these are gas bubbles, as indicated by

the smaller size and transparency of the spots (see red rectangles in Figure 4.1). In contrast, a few spots are opaque, larger, round, slightly milky, and yellow (see green squares *ibid.*). These are grown colonies. Das die einen Kolonien sind und die anderen Gasblasen ist streng genommen ja bereits eine Interpretation, finde es aber in dem Kontext recht nervig und wenig förderlich für die Verständlichkeit, das nur vage zu bezeichnen als milchig und weniger milchig. Wie siehst du das Max?

No mature colonies of *M. thaueri* were detected on the two plates. *M. boviskoreani*, on the other hand, grew on both plates. Individual colonies were picked from the plates, propagated in liquid DSM 1523, and verified as *M. boviskoreani* via 16S-Sequencing.

Pour plating

The pour plating was not successful for 100 µl of a full-grown *M. thaueri* culture with DSM 119 (1% agar?) as you can see in fig. A1. This experiment was also performed as duplicate.

Roll Plating

200 µl of full-grown *M. thaueri*, *M. wolinii* and *M. boviskoreani* culture have been roll plated in DSM 1523(1% agar?).

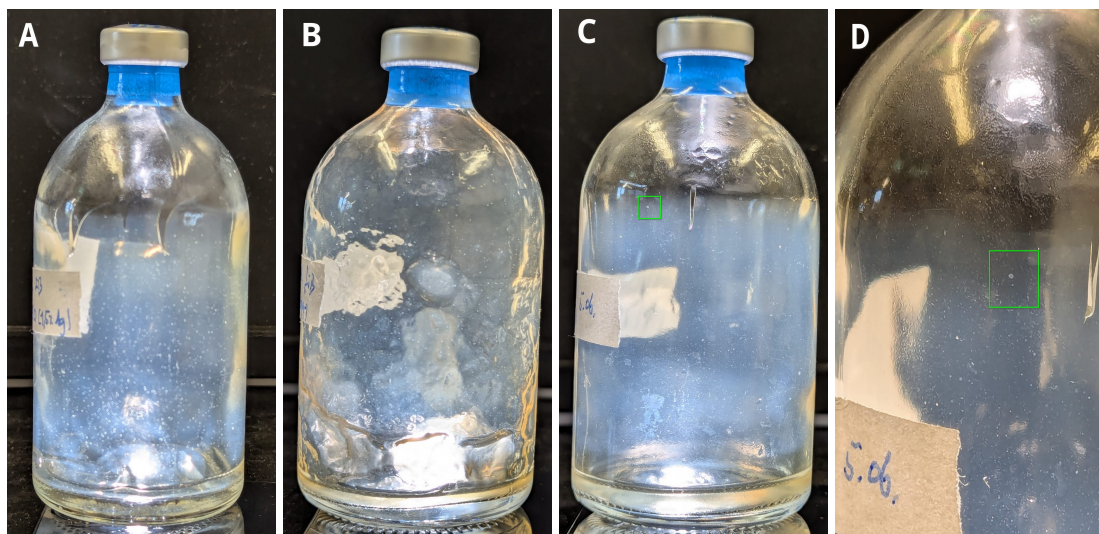


Figure 4.2: Roll plating of *Methanobrevibacter thaueri* CW^T (*M. thaueri*), *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). 200 µl of a full-grown culture were used. The green rectangle marks the same colony in C and D. A: *M. boviskoreani*, B: *M. wolinii*, C: *M. thaueri*, D: *M. thaueri* zoomed in on colony

In all three bottles CFU were formed (see fig. 4.2). *M. thaueri* showed notable less colonies (see table 4.2: 15 for 2432 or 3536), but they were larger than the little dots for the other cultures(fig. 4.2). Another difference between the bottle of *M. wolinii* (fig. 4.2 B) and the bottles of *M. thaueri* (*ibid.* C, D) and *M. boviskoreani* (*ibid.* A) is the agar structure. At *M. wolinii* it is

not as smooth and a bit more clumpy. In diskussion auf unterschiedliche technick beim rollen verweisen -> schlussfolgern wie am besten zu performen ist

However, besides the growing spots inside the bottles, shown in figure 4.2, also the H₂/CO₂ headspace changed. So approximately every two days the gas pressure in tbottle dropped from about 1.1 bar to below 0.8 bar and hat to be refilled. weist auf anaerobic life hin, ob die spots jetzt wirklich

Verification via 16S-Sequencing, as is done with spread plating, is still pending.

Even though the OD600 of the cultures used was not measured prior to plating, the plating efficiency was determined. This was done on the assumption that the cultures that appeared visually full-grown had an OD600 approximately equal to those listed in tab. 4.1, with the higher value for cell count per mL always being used.

$$\text{Plating efficiency [ppm]} = \frac{\text{counted CFU}}{\text{number of used cells}} \cdot 10^6 \quad (11)$$

The calculation was based on equation 11 and was performed as follows, for example:

$$\text{Plating efficiency} = \frac{2432}{0.2 \text{ ml} \cdot 10.819 \cdot 10^8 \text{ cells} \cdot \text{ml}^{-1}} \cdot 10^6 = 11.24 \text{ ppm}$$

Table 4.2: Counted colony forming unit (CFU) of different *Methanobrevibacter* (*M.*) culture for roll plating technic. 200 µl of a full-grown culture was used for plating. For *M. wolinii* and *M. boviskoreani* was counted a sixteenth of the bottle. Always the higher number of cells per ml from table 4.1 were used as a reference for the cell count in the initial solution.

culture	counted CFU	plating efficiency [ppm]
<i>M. wolinii</i>	2432	11.24
<i>M. thaueri</i>	15	0.13
<i>M. boviskoreani</i>	3536	19.07

4.2 lists the CFU counts per bottle as well as the corresponding plating efficiency. For *M. wolinii* and *M. boviskoreani*, the efficiency is in the lower two-digit ppm range. The efficiency for *M. thaueri* is notable lower, at 0.13 ppm.

4.1.4 Antibiotic sensitivity

Puromycin

M. thaueri and *M. boviskoreani* were cultivated in DSM 1523 media with various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. The experiment was performed in triplicates, with each antibiotic (AB) concentration series inoculated

with one full-grown pre-culture. The OD600 was taken once a day and plotted logarithmically against time in figures 4.3 and 4.4. For plotting negative measured OD600 values were set to 0.001. The original data and the adjusted OD600 values are in the appendix (see table B8 and B9). Max die tabellen sind halt extrem lang und gehen dann über drei oder vier seiten, sollen die dann trzdm einfach so in den Anhang? oder kann ich die bei github mit hochladen als csv

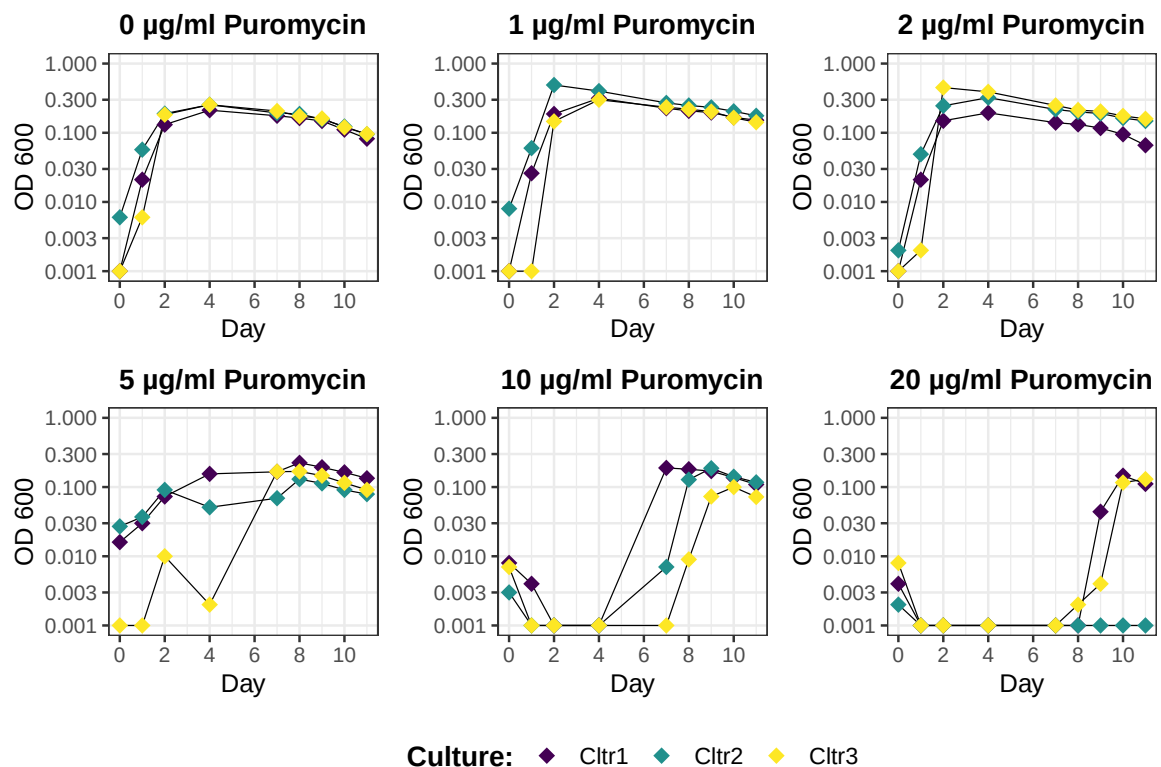


Figure 4.3: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) for various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. Experiment was performed as triplicates (Ctr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Fig. 4.3 shows that *M. boviskoreani* exhibits a very similar growth pattern for Puro concentrations of 0, 1, and 2 µg/ml. Exponential growth occurs within the first two days. This is followed by a delayed growth phase, such that the growth maximum is reached between days two and four at an OD600 of around 0.3. Until the final measurement day (day 10), the OD decreases continuously to an OD600 value of around 0.1. One difference that can still be noted here, however, is that the variation between the triplicates at 0 µg/ml is smaller than that observed with Puro. In the latter case, culture 3 does not begin exponential growth until one day later

as the other triplicates.

At 5 µg/ml, the growth curves for Cltr. 1 and Cltr. 2 are generally similar to those just described. However, the two cultures do not grow as quickly. Cltr. 3 shows delayed growth and exhibits greater fluctuations in the OD600 values. Thus, all three triplicates here reach their maximum growth only after 8 days. Nevertheless, the maximum OD600 values here are also slightly lower than in the previous growth curves. Thereafter, the OD600 values decrease continuously.

At 10 and 20 µg/ml Puro concentrations, exponential growth begins with a notable delay. At 5 µg/ml, this occurs between 4 and 7 days. The maximum is reached between 7 and 10 days, with maximum OD600 values ranging from approximately 0.1 to 0.25. At 20 µg/ml, one replicate shows no growth at all by the end of the measurement (Cltr. 2). The other two replicates do not begin exponential growth until day 8 or 9. The maximum is reached on day 10 at approximately 0.1. By day 11, a slight decrease in the OD600 values can already be observed.

The growth curves of *M. thaueri* for Puro concentrations of 0 and 1 µg/ml reach their OD600 maximum after 3 days, similar to *M. boviskoreani* (see Figure 4.4). However, the maximum does not exceed 0.3 and is closer to 0.2 than to 0.3.

After that, the OD600 decreases rapidly again, so that at 0 µg/ml, Clt. 1 and 2 already have an OD below 0.003 after 9 days, and Cltr. 3 is also just barely above 0.01 after 10 days. For 1 µg/ml, the final measured OD values range between 0.001 and 0.05.

For 2 µg/ml puromycin, Figure 4.4 shows growth for all three triplicates, reaching its maximum on day 3. However, the increase in OD600 values is significantly lower. Thus, the maximum for Cltr. 2 is below 0.03, and for the others it is around 0.1. This is followed by another decrease in the values, so that the final values range between 0.001 and 0.03.

For the other measured Puro concentrations, in contrast to *M. boviskoreani*, no growth was observed with one exception (Cltr. 3 at 5 µg/ml) (see Fig. 4.3 and 4.4), with the OD600 values remaining below 0.003. For Cltr. 3 at a Puro concentration of 5 µg/ml, an OD600 of more than 0.1 was measured on day 3, but this value dropped back to 0.003 by day 6.

In der Diskussion drauf verweisen, dass man aus der Kurve für thau auch die Lyse gut erkennt, die auch bei den Beobachtungen bei der Kultivierung mit benannt wurden.

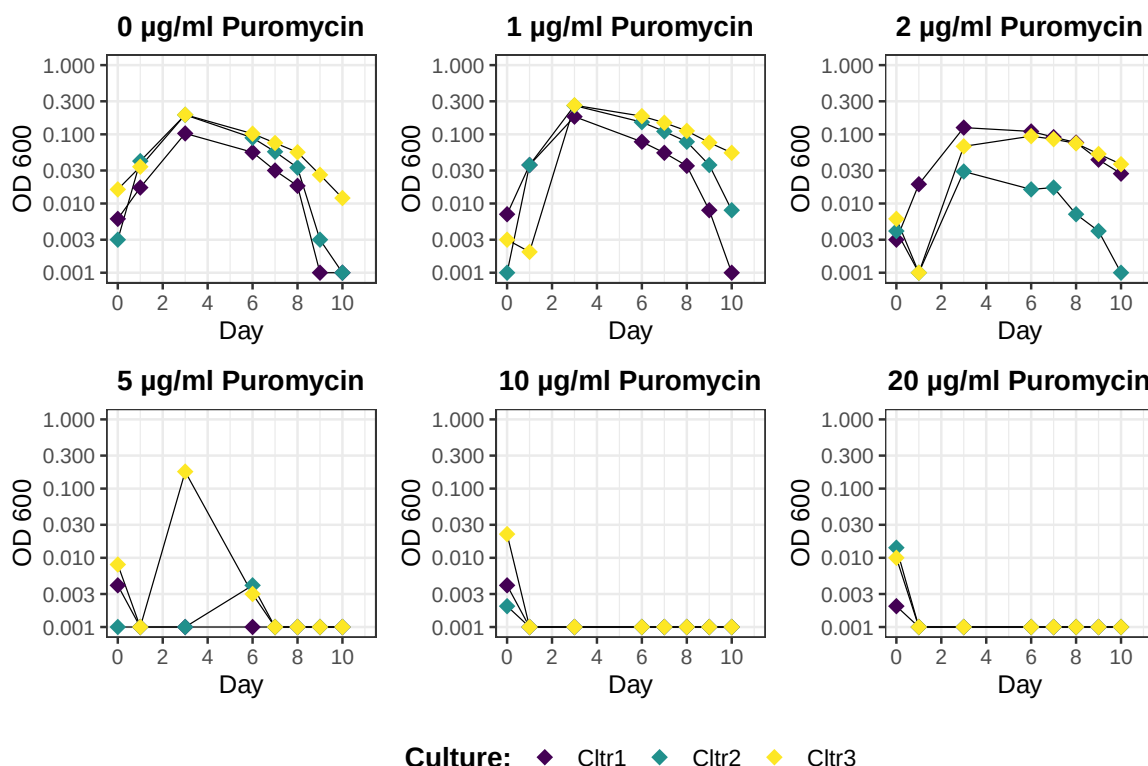


Figure 4.4: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) for various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. Experiment was performed as triplicates (Ctrl. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Cycloserin

In duplicates it was tested if 200 µg/ml cycloserin (Cyc-Se.) would inhibit the growth of *M. thaueri* in DSM 1523. In all four bottles (two negativ and two with 200 µg/ml Cyc-Se.) visual growth could be detected

4.1.5 Nucleotide-baseanalogica sensitivity test

8-Azahypoxanthin

Similar to the AB sensitivity test, a growth experiment was conducted for the base analogs 8-Azahypoxanthin (8-Aza) for *M. thaueri*, *M. boviskoreani*, *M. millerae*, and 6-Azaauracil (6-Aza) for *M. wolinii*. One difference is that the base analogs do not dissolve in H₂O, so they were dissolved in dimethylsulfoxid (DMSO). Furthermore, different concentrations were selected: 0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml, and one with 1 ml DMSO. The OD600 was measured and plotted on a logarithmic scale against time (see Figures 4.5

through 4.8). The original measured data is in the appendix (Fig. X).

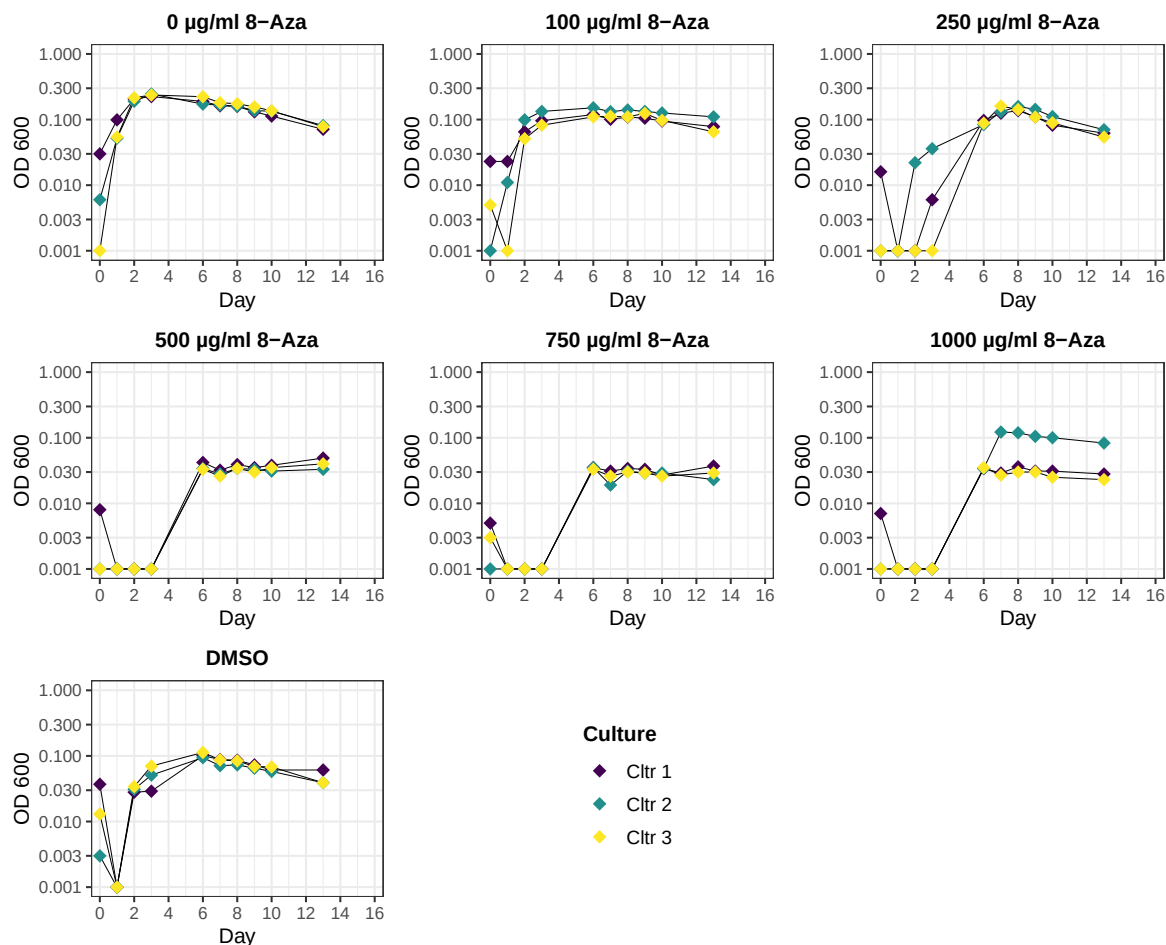


Figure 4.5: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

4.5 shows a growth curve for the positive control (0 µg/ml 8-Aza) of *M. boviskoreani* that is similar to that of the positive control in Fig. 4.3. Thus, exponential growth occurs within the first two days. On days 3 through 4, the OD600 reaches a maximum of around 0.3, followed by a continuous decline with a final OD600 of just under 0.1 on day 13.

The solvent control (DSMO), in which no base analogs were added but only 1 ml of the solvent DMSO, showed a different growth curve. Initially, none of the triplicates begin to grow immediately; instead, the OD600 values first decrease. On day 3, slight growth begins, reaching a maximum of 0.1 on days 4–6. Afterward, the values decline to approximately 0.03 on day 13.

At the lower concentrations of 8-Aza (100 and 250 µg/ml), *M. boviskoreani* in fig. 4.5 a growth curve similar to that of the corresponding positive control. However, at 100 µg/ml 8-Aza, the maximum OD600 value is reached within 1 to 3 days, but it is approximately 0.1 instead of just under 0.3. 0.1 is also the maximum OD600 value for the 250 µg/ml concentration; here, growth does not begin until 2 to 4 days later, so that the maximum is not reached until 7 days later.

The three remaining 8-Aza concentration solutions (500, 750, and 1000 µg/ml) exhibit a very similar growth curve. An OD600 below 0.01 was measured for *M. boviskoreani* during the first 4 to 6 days. Then there was an increase to an OD600 of 0.03, which remained constant until the final measurement day (day 13). An exception is Culture 2 in the 1000 µg/ml 8-Aza culture. This culture grows from Day 3 to Day 5 to an OD600 of 0.1, which then decreases slightly until the end of the measurement series.

Figure 4.6 visualizes the growth curves for *M. thaueri*. The first thing that stands out in comparison to the other archaea (fig. 4.5, 4.7, and 4.8) is that the baseline OD600 is higher, at approximately 0.03. With the exception of the positive control (0 µg/ml 8-Aza), no significant increase in the OD600 values was observed. Instead, there was a fairly constant fluctuation around the initial value of 0.03 until the last measurement day (Day 16). An exception here is the triplicate Cltr. 3 in the DMSO sample. Here, an OD600 of 0.3 was measured on Day 2, but it fell back to 0.03 by the next measurement day (Day 4). At the end of the measurement series, this triplicate also ended at approximately 0.07 instead of 0.03 like the other triplicates.

In the positive control, slight growth of *M. thaueri* was detected. However, the growth curve differs quite noticeably from that of the positive control for *M. thaueri* in fig. 4.4. In addition, there is considerable heterogeneity among the individual triplicates.

For example, Cltr. 1 begins to grow slightly from the first day and reaches its maximum OD600 between 0.1 and 0.3 on day 9 (see fig. 4.6). By the end of the measurement, the OD600 decreases continuously to 0.1.

Cltr. 3 begins to grow on Day 8, reaching a maximum OD600 of approximately 0.2 on Day 12. Between Days 13 and 14, the last triplicate then begins to show slight growth, reaching an OD600 of just over 0.1 on Day 16.

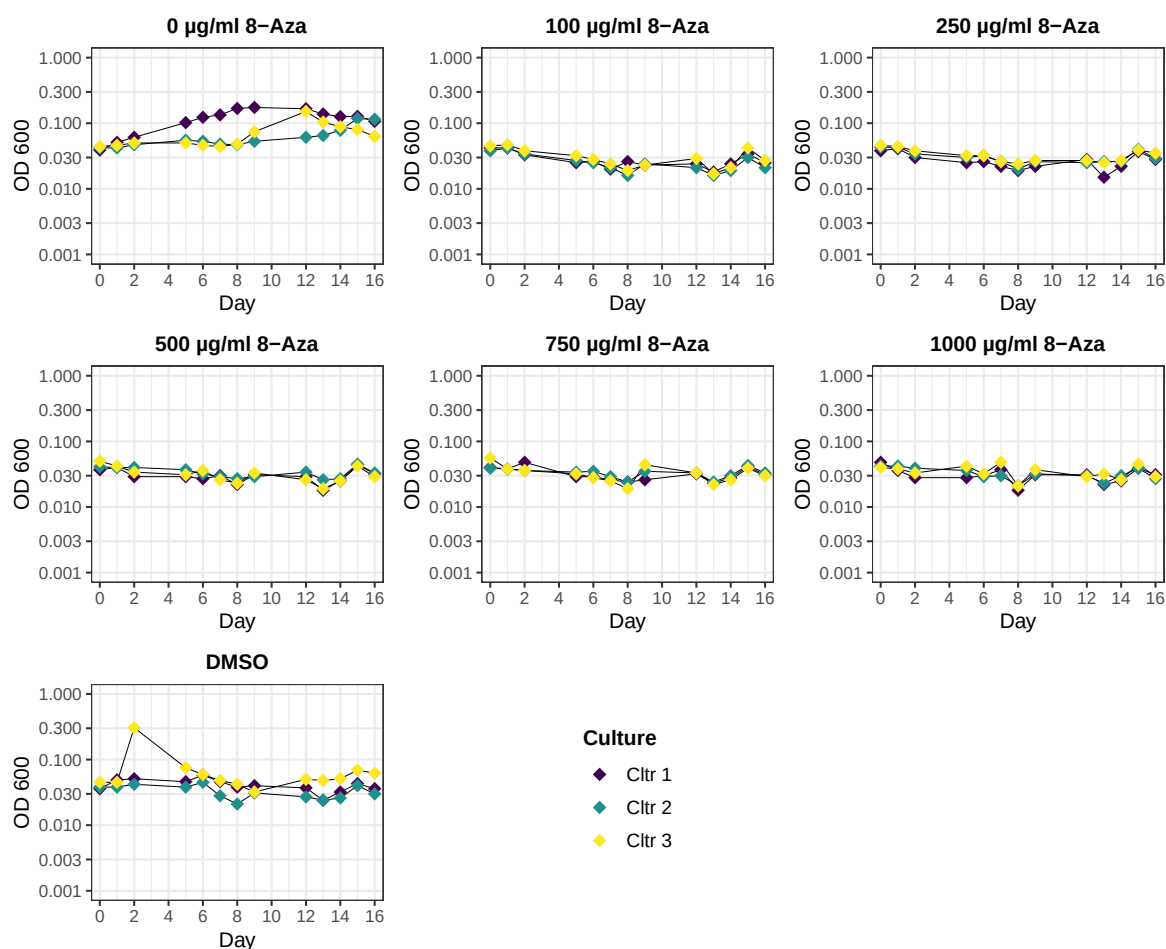


Figure 4.6: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

The growth curves of *M. millerae* for the positive control (0 µg/ml 8-Aza) and the solvent control (DMSO) have a similar shape for all triplicates, with the measured OD600 being reached on day three, followed by a continuous decline until the end of the measurement series (day 14). However, the curves differ in their OD600 values. The positive control reaches a maximum value of 0.3, while the solvent control reaches just under 0.1. This also results in a significantly lower final OD600 measurement for DMSO (below 0.03) and for the positive control (just under 0.1).

Clear trends for the other samples are generally difficult to discern at first glance; however, most triplicates do not reach an OD600 higher than 0.1.

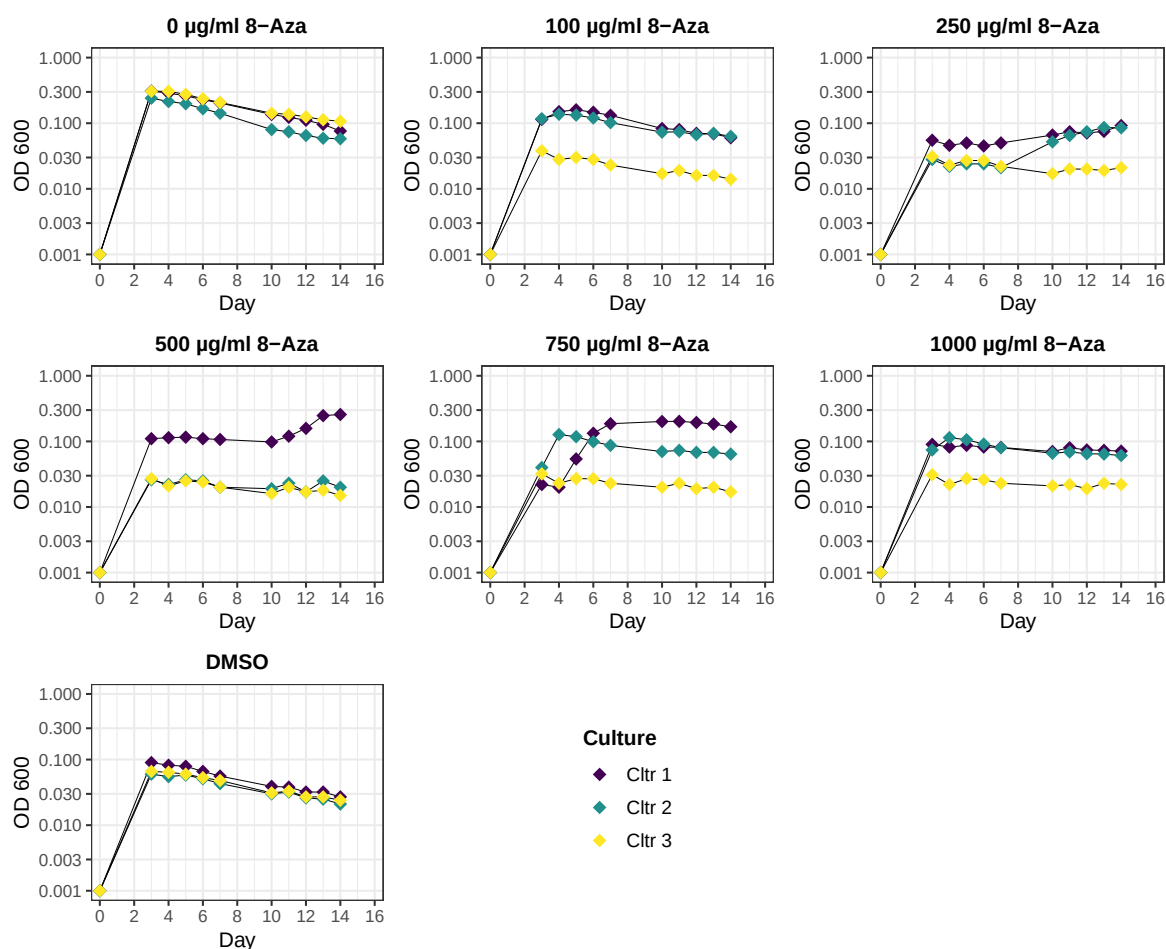


Figure 4.7: Growthcurve in DSM 1523 for *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

The 100 µg/ml 8-Aza generally follows the curve shape of the positive and solvent controls, although triplicates Cltr. 1 and 2 reach their maximum OD600 value on Day 4 with a value above 0.1, while Cltr. 3 does not exceed 0.03.

For the 250 µg/ml concentration, the triplicates have a constant OD600 of approximately 0.03, with Cltr. 1 and 2 showing slight growth starting on day 7, reaching approximately 0.1 by the end of the measurement. The triplicates for Ctr. 2 and 3 at the 500 µg/ml concentration show a constant OD600 value of 0.03 starting on Day 3. For Ctr. 1, an OD600 of 0.1 was measured between Days 2 and 10, which increases to just under 0.3 by Day 13.

For the 750 µg/ml and 1000 µg/ml 8-Aza solutions, replicate Ctrl. 3 exhibits a constant value of 0.03 starting on day 2. At 1000 µg/ml, the other two triplicates show a constant OD600 of just under 0.1 starting on day 2, while at 750 µg/ml, only Cltr. 2 shows this value starting on

day 2. The Cltr. 1 triplicate in the 750 µg/ml solution shows a value of 0.03 on days 2 and 3, then increases to an OD600 between 0.2 and 0.3 by day 7. This value decreases slightly by the end of the measurement series.

6-Azaauracil

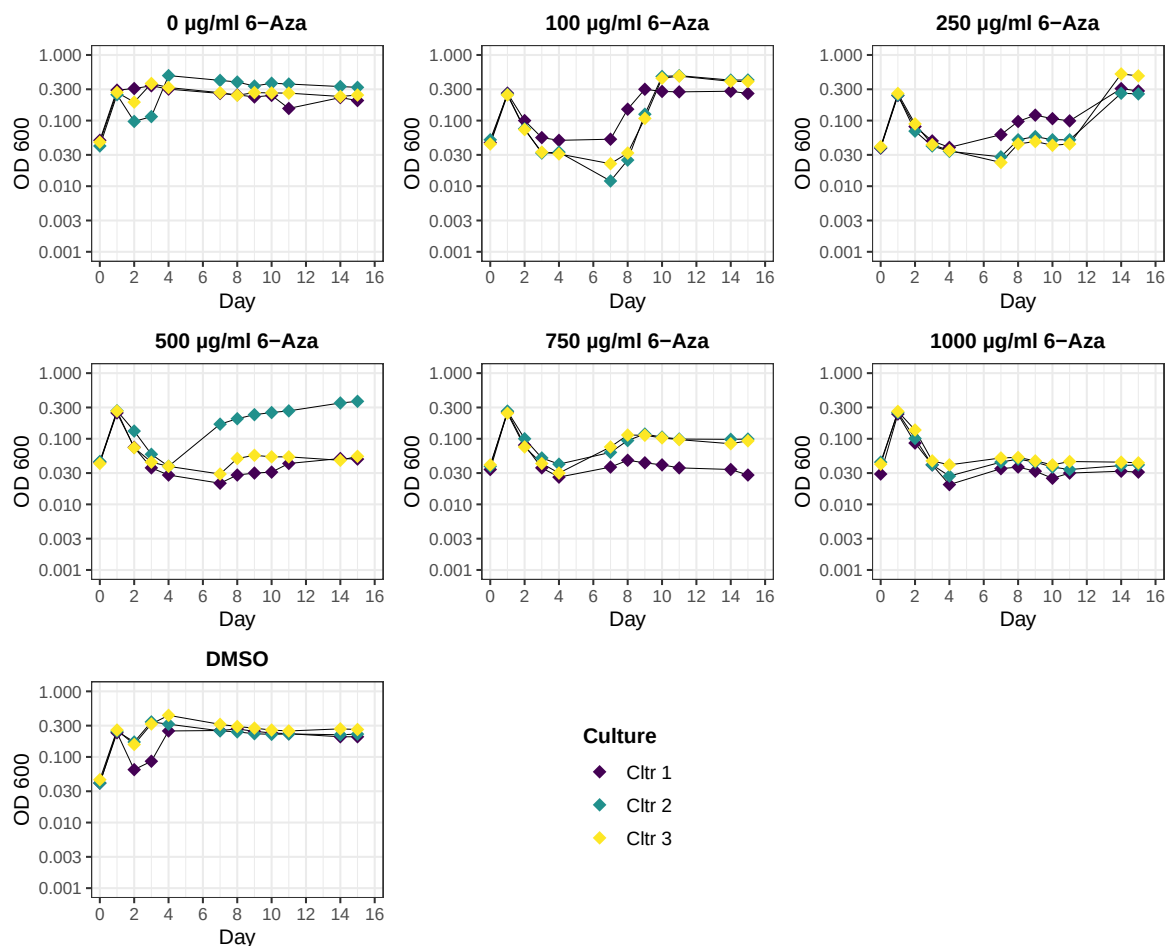


Figure 4.8: Growthcurve in DSM 1523 for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) for various concentration of 6-Azaauracil (6-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 6-Aza. Experiment was performed as triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

All growth curves for the 6-Azaauracil (6-Aza) test series by *M. wolinii* (see fig. 4.8) show a jump in the OD600 values within the first day from an initial value of approximately 0.03 to 0.3, followed by a drop that varies in magnitude depending on the sample and results in different curve profiles.

The triplicate Cltr. 1 of the positive control (0 µg/ml) shows no change after Day 1 and main-

tains a constant OD600 value of around 0.3 until the end of the measurement series. Cltr. 3 dips slightly to approximately 0.2 on Day 2, then remains constant at 0.3 starting on Day 3. The triplicate Cltr. 2 shows the greatest change after Day 1, remaining at a value of 0.1 on measurement days 2 and 3, before rising to an OD600 of approximately 0.4 on Day 4. The final measured value for this triplicate on Day 15 is approximately 0.3.

The triplicates in the solvent control (DMSO in fig. 4.8) show a more pronounced drop in OD600 values than the positive control. For example, Cltr. exhibits an OD600 of slightly more than 0.1 on Day 2, while Cltr 1 drops to a value below 0.1 on Days 2 and 3. All three triplicates have a fairly stable OD600 of approximately 0.3 starting on Day 4.

In all 6-Aza preparations, a decrease in OD600 values was observed down to a value below approximately 0.03. All triplicates in the 100 µg/ml preparation grew to an OD600 of 0.3 between measurement days 7 and 10, which remained constant until the end of the measurement series.

In the 250 µg/ml preparation, significant growth to an OD600 of more than 0.3 was not observed until between days 11 and 14, even though all triplicates showed a slight increase between days 7 and 8 of approximately 0.03, so that Cltr. 1 had a value of 0.1 and the other two triplicates had values of approximately 0.07.

No further growth was observed for any of the triplicates of the 1000 µg/ml concentration by the end of the measurement series. The same was true for triplicates Cltr. 1 and 3 of the 500 µg/ml 6-Aza concentration and Cltr. 1 at 750 mg/ml.

The OD600 of Cltr 2 in the 500 µg/ml preparation rises from 0.03 on measurement day 4 to approximately 0.2 on measurement day 7, and subsequently to a final OD600 of more than 0.3 by the end of the measurement series.

DMSO

In general, all preparation with DMSO showed the tendency to be more vulnerable to turn red when taking a sample for photometric measurement. In addition, The decolorization also took longer than in the positive control or other experimental conditions without DMSO, sometimes over an hour.

This interaction of DMSO with the DSM 1523 media was investigated more closely in the *M. thaueri* measurement series. Here, one triplicate with DSM 1523 was prepared with just

1 ml DMSO, but without inoculation (DMSO –, see fig. 4.9).

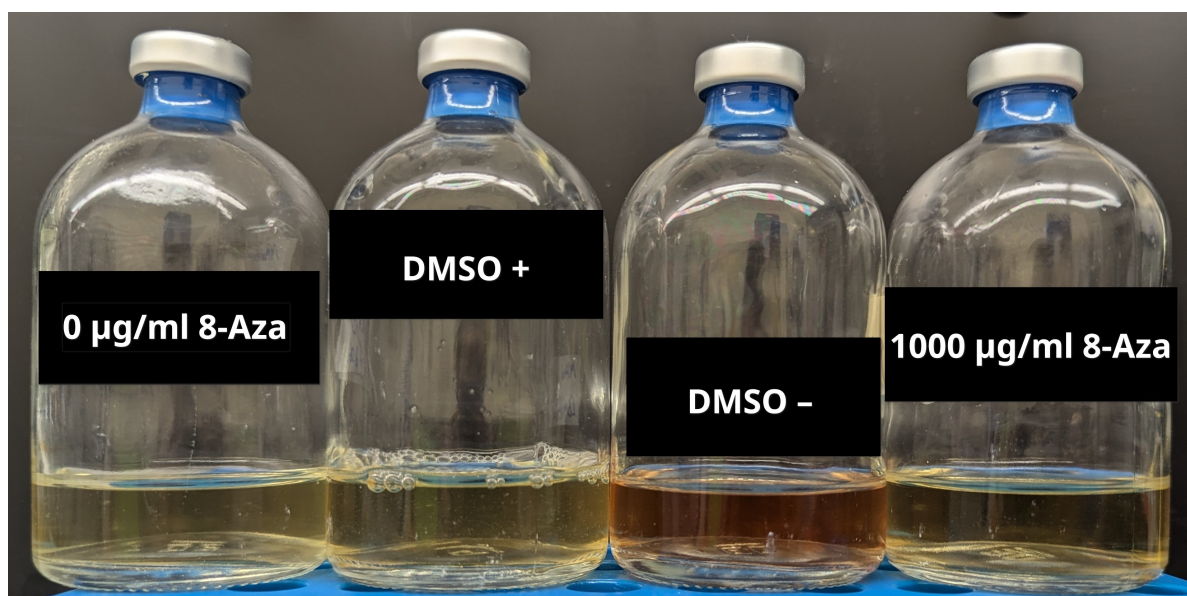


Figure 4.9: Color differences with and without DMSO after 6 days of incubation. Image of various preparations for *M. thaueri* from the same triplicate (Cltr. 3). The positive control (0 µg/ml 8-Azahypoxanthin (8-Aza)) contains only the organism (*M. thaueri*) in DSM 1523, 1000 µg/ml 8-Aza *M. thaueri* and the named base analog, DMSO + *M. thaueri* and 1 ml DMSO; DMSO – also contains 1 ml DMSO but was not inoculated.

After the first few days of incubation, a difference in color was observable among the cultures. Specifically, the DMSO bottles exhibited a red color, in contrast to the yellow color of the other samples in the 8-Azahypoxanthin (8-Aza) measurement series for *M. thaueri*. This is illustrated in fig. 4.9 using selected samples from the triplicate Cltr. 3 after 6 days of incubation. The OD600 was also measured for DMSO, but showed no difference from the baseline OD600 values of the other samples (approximately 0.03 to 0.04).

4.1.6 Plasmid designing

***pac*-gene**

The *pac*-gene in the initial plasmid (*pac_old*), that should have been taken for the transformation via conjugation of the archaea, showed a different aminoacids (AA) sequence to the original sequence (*pac_org*) from *Streptomyces alboniger* (see (a) in fig. 4.10). They differ in five AA on the C-terminal end.

These mutation led in to different 3-D structures. In (b) of fig. 4.10 the original *pac* has an antiparallel β -sheet with a big loop on the right side of its structure. The *pac_old* ((c)) shows also an antiparallel β -sheet at this position, but it is longer and has a shorter loop.

Plasmid

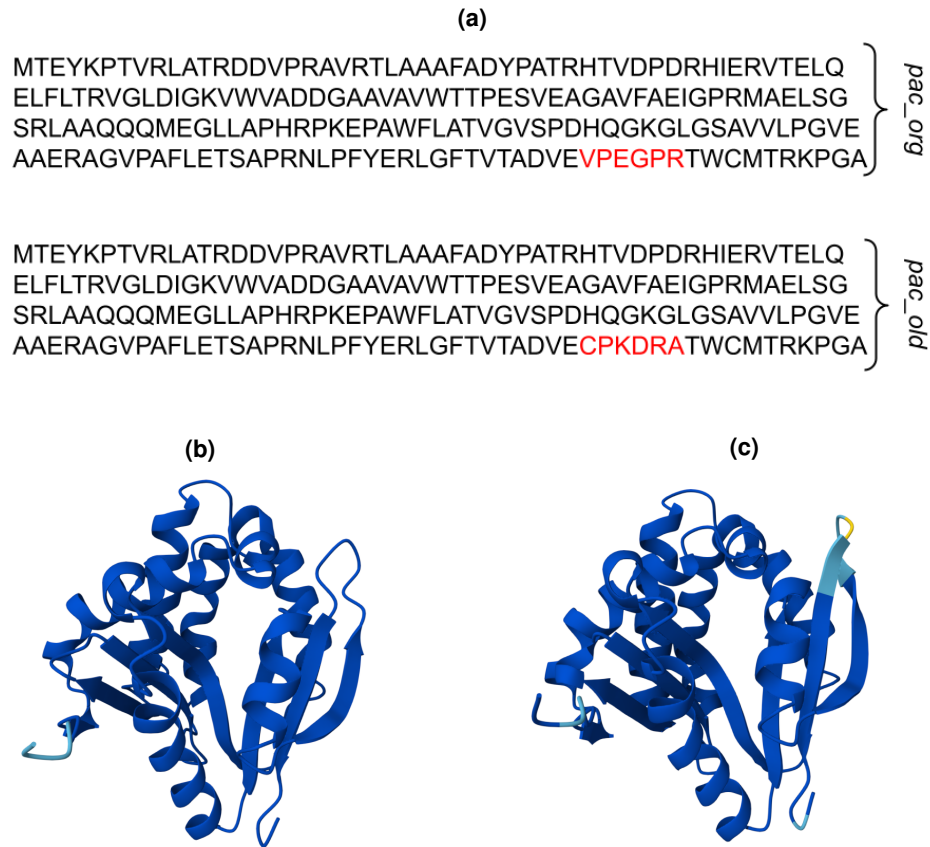


Figure 4.10: Aminoacids (AA) sequence of two *pac* genes and their 3-D structures. (a) shows the AA sequence of the original sequence from *Streptomyces alboniger* (*pac_org*) as well as from the *pac* gene that was located in the plasmid originally intended for transformation (*pac_old*). The differing section is highlighted in red. (b) The projected 3D structure for *pac_org* calculated by AlphaFold. (c) The projected 3D structure for *pac_old* calculated by AlphaFold.

In SnapGene, four plasmid maps were created for *M. thaueri* and two for *M. boviskoreani*. The constructs followed the general scheme shown in fig. 1.1 and are listed in ???. For the plasmids, the original *pac* sequence (*pac_opt*) was codon-optimized for the respective archaea.

Two plasmids were constructed, as shown in fig. 4.11: pMt_0231_hpt for *M. thaueri* and pMb_0231_hpt for *M. boviskoreani*. Compared to the plasmid originally used for the construction, a new restrictionsite (SpeI) was inserted between the *pac* and its promoter (see fig. 4.11). In addition, the plasmid *pac_old*, which contained the alternative AA sequence, was converted to the codon-optimized version with the original AA sequence (*pac_opt*), and the promoter was changed to hmtB.

Both plasmids were verified by sequencing.

Table 4.3: Suicide plasmid constructs for transformation of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) via conjugation. The listed plasmids follow the general structure shown in fig. 1.1, with a cassette for *E. coli* containing the conjugation gene and homologous arms for recombination. The name of each plasmid is derived from the organism for which it possesses the appropriate homology arms (pMt for *M. thaueri* and pMb for *M. boviskoreani*) and the four named columns, where ORI refers to *M.* in this case and the promoter for the *pac*. *pac_org* is the *pac* with the original sequence derived from *Streptomyces alboniger*, *pac_opt* is the codon optimized version for the affected organism. The size of the constructed is given in bp as well.

name	ORI	promoter	pac	extra feature	size (bp)
pMt_0231_hpt	NO (0)	hmtB (2)	pac_opt (3)	hpt (1)	6145
pMt_0331_hpt	NO (0)	PglnA (3)	pac_opt (3)	hpt (1)	6254
pMt_0241_hpt	NO (0)	hmtB (2)	pac_org (4)	hpt (1)	6145
pMt_0341_hpt	NO (0)	PglnA (3)	pac_org (4)	hpt (1)	6254
pMb_0231_hpt	NO (0)	hmtB (2)	pac_opt (3)	hpt (1)	6214
pMb_0331_hpt	NO (0)	PglnA (3)	pac_opt (3)	hpt (1)	6323

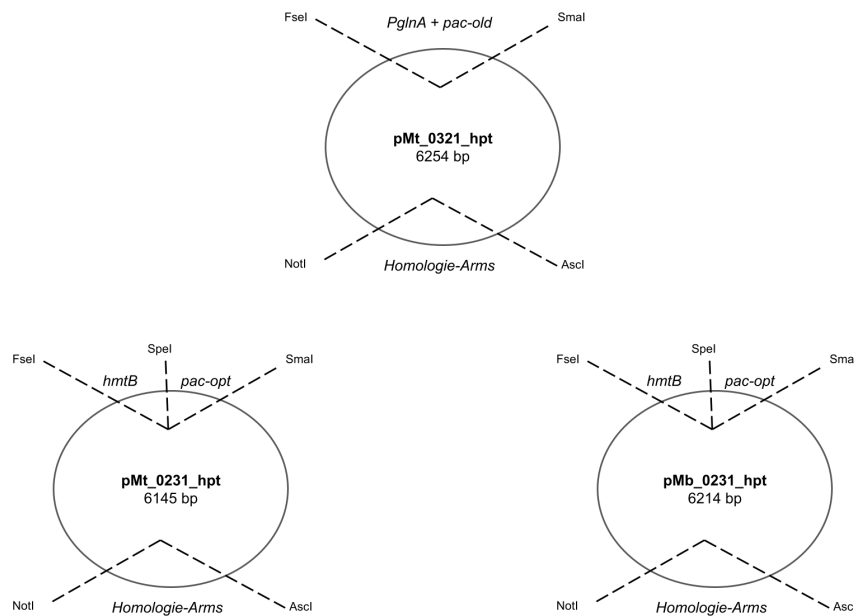


Figure 4.11: Suicide plasmid maps for transformation by conjugation for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). pMt_0321_hpt is the parent plasmid used to construct the other two. The maps show only the gene regions that have been modified, as well as the restriction site sites used or the newly created restriction site sites.

4.1.7 Transformation via conjugation

E. coli cultivation test in DSM 1523

Cultivation of *E. coli* DH5 α was tested under similar settings in DSM 1523 as for *Methanobrevibacter*. Three drops for inoculation was enough to get a dense culture the next day.

Preparation for transformation

Hier erst in Leipzig text möglich, weil da das laborbuch liegt -> erstelle Tabelle mit Ansätzen und dann welche erfolgreich waren
+ Sequenzergebnisse -> bestätigung von bovis ja vorhanden -> komplettes genom is still pending

4.2 Geography

All methane emission analyses focused on three countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). In addition, only the anthropogenic emission will be investigated.

4.2.1 Total methane emissions of a country

Figure 4.12 shows the development of the total methane emission of a year as well as of the specific sectors from 1990 to 2020s for each country.

In general, the first thing that stands out is that the time series for the U.S. ends as early as 2022, while those for the other two countries do not end until 2024. Furthermore, the scale of methane emissions shows that the U.S. generally produces significantly more CH₄; the maximum total emissions are 35,000 kt CH₄, while for DEU the maximum is 5 Tg and for NZL it is 1.5 Tg.

For Germany (DEU), a clear downward trend is evident in total CH₄ emissions (see fig. 4.12). Emissions fell from 3 Tg CH₄ in 1990 (5 Tg) to 2 Tg in 2024, representing a decrease of approximately 60%. Two distinct phases of this trend can be identified: first, from 1990 to 2006, with a sharp decline of about 2.5 Tg CH₄ (approximately 2.6 Tg CH₄ in 2006), followed by a phase with a more moderate decline to 2 Tg of anthropogenically produced methane in 2024.

Total emissions excluding those from enteric fermentation (EnterFe) in cattle (total excluding cattle EnterFe) follow a similar trend, with the curve's value being only 1 Tg CH₄ lower.

Looking at the sectoral emissions of DEU, it becomes apparent that in 1990 there were three equally large CH₄-producing sectors: "Energy," "Waste," and "Agriculture," each accounting for approximately 1.5 Tg CH₄ (see fig. 4.12). However, these sectors have developed differently. The energy and waste sectors have followed quite similar trends. Following a sharp decline between 1990 and 2006—with the waste sector not beginning to change until 1994—from 1.5

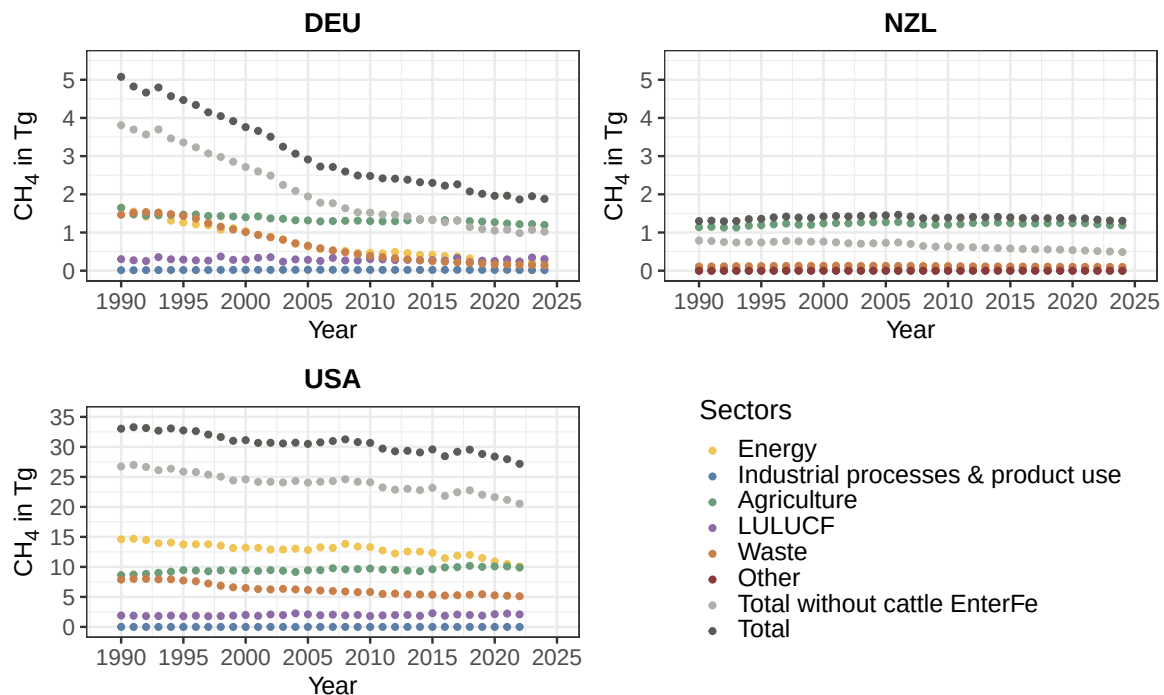


Figure 4.12: Time series of the anthropogenic methane emission of Germany (DEU), New Zealand (NZL) and United States of America (USA). The different relevant emission sectors are shown with their total sum (Total). Furthermore, the total emission minus the CH₄ produced in enteric fermentation (EnterFe) processes of cattle is plotted as well. Methane is plotted as Tg CH₄ of a year against the time.

to 0.5 Tg, a moderate decline then sets in, until approximately 0.025 Tg of CH₄ is emitted in 2024.

Agriculture is also changing, though only very slightly compared to the other two sectors: from 1.6 Tg of methane in 1990 to approximately 1.25 Tg CH₄ in 2024, with the largest single change occurring between 1990 and 1991, amounting to a difference of about 0.1 Tg. In 2015, total emissions excluding cattle EnterFe are exactly as high as those of the agricultural sector and fall below that level thereafter.

In addition to these three sectors, two others are shown in fig. 4.12.

The “Industrial Processes & Product Use” sector is a negligibly small source of methane emissions. Its CH₄ emissions have remained consistently well below 0.1 Tg over the 35-year period.

With CH₄ emissions consistently ranging between 0.3 and 0.4 Tg, the land use and land use change and forestry (LULUCF) sector was the fourth-largest single methane emission sector in Germany during the 1990s, but it surpassed the waste sector in 2016 and has been the second-largest CH₄-emitting sector in DEU after agriculture since 2019.

For the USA the difference between total methane emissions and total emissions excluding cattle EnterFe remains fairly constant at around 10 to 15 Tg CH₄ (see fig. 4.12). In addition, a clear, steady decline can be observed for the total methane emission from 1990 (approx. 33 Tg CH₄) to 2022 (approx. 27 Tg CH₄), although the curve does not follow the trend linearly but rather fluctuates slightly. Over this 32-year period, the U.S. has reduced its methane emissions across all sectors by about 6 Tg CH₄, which corresponds to a reduction of 18%.

As in Germany, the three most significant methane-producing sectors in the U.S. are “Energy,” “Agriculture,” and “Waste” in 1990; however, here the energy sector is the largest contributor, accounting for nearly half of the emissions in 1990 with 15 Tg CH₄, followed by the waste and agriculture sectors with approximately 8 Tg CH₄ each.

The energy and waste sectors show a downward trend over the time series (energy: from 15 to 10 Tg CH₄ and waste: from 8 to 5 Tg CH₄), while emissions from agriculture rise slightly over time, from 8.5 Tg to 10 Tg CH₄. Thus, in 2022, agriculture and energy—each at 10 Tg CH₄—are the largest methane emission sectors and together account for more than two-thirds of total methane emissions in the U.S. in 2022.

Two other sectors contribute to U.S. methane emissions: the “Industrial Processes & Product Use” sector, which is relatively small and therefore negligible, and the “LULUCF” sector, with a constant range of 0.2 to 0.25 Tg CH₄.

Compared to the other two countries, New Zealand presents a different picture (see fig. 4.12). First, total CH₄ emissions in 2024 are nearly the same as those in 1990, namely about 1.4 Tg CH₄; furthermore, a slight increase in emissions to 1.5 Tg CH₄ can be observed through 2006, followed by a slight decline.

Second, “agriculture” is the dominant sector in terms of methane emissions and accounts for nearly 90% of CH₄ emissions from NZL (see fig. 4.12). All other sectors account for significantly less than 0.1 Tg CH₄ per year, with the exception of the waste sector, whose methane emissions are slightly above 0.1 Tg CH₄.

Third, the difference between total methane emissions and total emissions excluding cattle EnterFe increases by about 60% (0.5 Tg CH₄ in 1990 and 0.8 Tg in 2024).

4.2.2 Enteric fermentation processes of cattle

Methane emissions

Figure 4.13 shows the trend in methane emissions from enteric fermentation in cattle, as well

as their subcategories “dairy cattle” and “non-dairy cattle” for each country. The values for the subcategories are calculated according to eq. (1) and based on the required data from the reported CRT (visualized in fig. A2 through A4). Total emissions from cattle are therefore simply the sum of these values.

Cattle emissions for Germany (DEU) in fig. 4.13 decline from 1990 to 2024 (from 1.25 Tg CH₄ to approximately 0.8 Tg), with a steeper decline in the first year of the time series, followed by a more gradual decline.

For dairy and non-dairy cattle, this initial decline is also evident in fig. 4.13, after which a slight decline can also be observed (dairy cattle: 0.75 Tg CH₄ in 1990 to 0.5 Tg in 2024; non-dairy cattle: 0.55 in 1990 to 0.35 in 2024). Dairy cattle are an exception: their emissions drop to 0.5 Tg CH₄ by 2006, then rise slightly again through 2015, and subsequently decline once more, returning to 0.5 Tg by 2014.

Dairy cattle consistently produce slightly more CH₄ per year than non-dairy cattle (0.1–0.2 Tg CH₄, see fig. 4.13. This contrasts with the herd ratio shown in fig. A2, where the dairy cattle herd generally accounts for about half of the non-dairy cattle herd over the years.

Further differences in the indicators are evident for Y_m, where dairy cattle had a significantly higher value of 7% in 1990 compared to 6.25, which, however, declines sharply to 6.1% by 2024, while the value for non-dairy cattle rises to 6.5%, as well as for GE, where the value for the non-dairy cattle herd remains constant at 100 MJ head⁻¹ day⁻¹, while the value for the dairy cattle herd rises from 250 to 380 MJ head⁻¹ day⁻¹.

Methane emissions from cattle in EnterFe for New Zealand rise from 0.5 to 0.75 Tg CH₄, while the values have remained relatively constant since 2014.

An increase is also evident for dairy cattle, which has plateaued since 2014 (0.25 Tg CH₄ in 1990 to 0.55 Tg in 2024). Non-dairy cattle, on the other hand, show similar values of around 0.5 Tg CH₄ in both 1990 and 2024.

Thus, in 1990, dairy and non-dairy cattle contributed equally to methane emissions from the cattle population; however, this share has changed over the years for dairy cows, so that by 2024 they will account for about two-thirds of the emissions. This shift in the ratio is quite similar across the subcategory populations (see fig. A3). Furthermore, the energy efficiency (GE) values for non-dairy cattle remain constant at 150 MJ head⁻¹ day⁻¹, while the value for dairy cattle rises from 160 to 225 MJ head⁻¹ day⁻¹.

In the USA, methane emissions from dairy cattle rise from just under 2 Tg CH₄ to 2 Tg, with the slight increase not beginning until around 2005. The other two livestock categories, on the other hand, fluctuate overall—for both non-dairy and dairy herds—between 4.9 and 5.5 Tg for non-dairy livestock and between 6.5 Tg CH₄ and 7.1 Tg for dairy livestock (see fig. 4.13). Unlike in the other two countries, non-dairy cattle here emit significantly more than dairy cattle (2024: 5 Tg CH₄ versus 2 Tg).

As with the other countries, the emission curves correspond fairly well with the livestock numbers from fig. A4, where non-dairy cattle—at 80 million head—are kept in significantly greater numbers than dairy cattle (20 million head). The curve for dairy cattle does not match the livestock numbers exactly, as these remain constant. However, the GE for dairy cattle rises steadily from 200 to 260 MJ/animal·day between 1990 and 2022, while the Y_m value decreases from 6.5% to 6.1% over the same period.

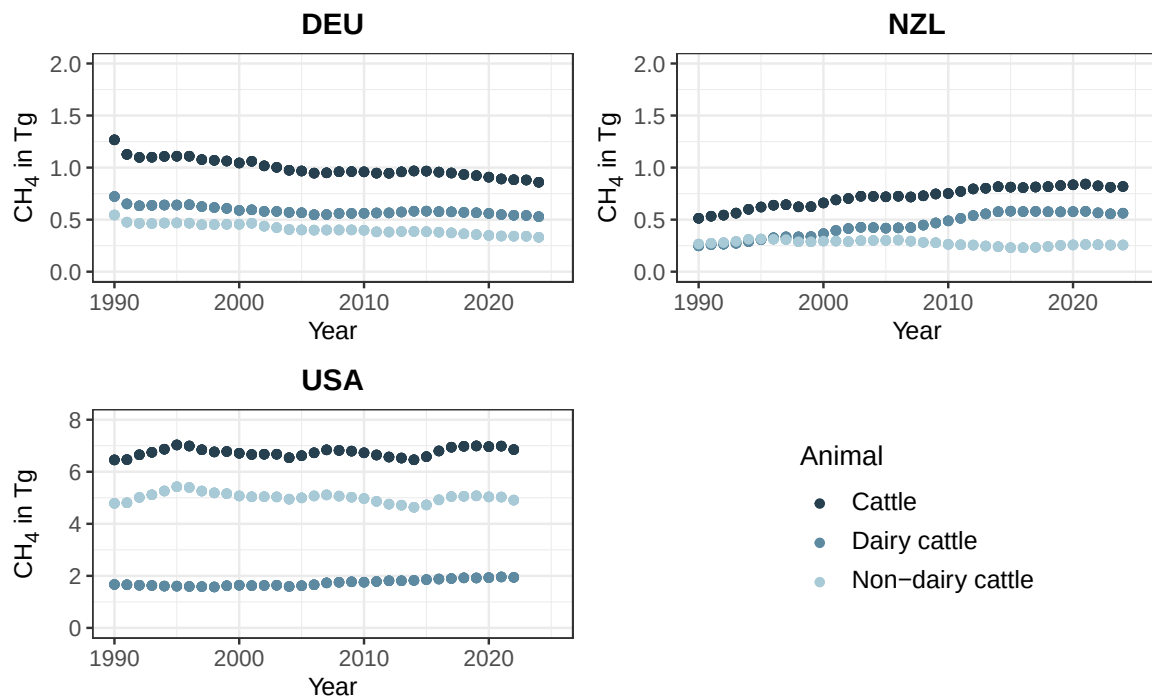


Figure 4.13: Time series of methane emission derived from enteric fermentation processes in cattle and its subcategories dairy and non-dairy for the countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). Methane emission are measured in emitted Tg CH₄. Emissions are calculated for dairy and non-dairy cattle with the indicators in fig. A2:A4 and the eq. (1). The cattle emission is than the sum of its subcategories.

When comparing cattle emissions across countries in fig. 4.13, it is striking that the U.S. produced more than three times as much in 2022 as the other two countries combined (6.8 Tg CH₄ compared to less than 2 Tg combined). This is also reflected in the number

of animals: in 2022, the US had approximately 100 million head of cattle, while DEU and NZL each had around 10 million (see fig. A2:A4).

Germany and New Zealand produce similar amounts of methane from cattle farming in 2024 (approx. 0.8 Tg CH₄), in 1990, DEU had produced twice as much, but also kept twice as many cattle (approx. 20 million animals in 1990 compared to 10 million in 2024).

Share of CH₄ emissions from EnterFe of cattle

The share of CH₄ produced through EnterFe processes in cattle on entire amount of emitted methane for the agriculture sector of a country declines slightly for Germany (approx. from 78% 1990 to 76% 2024, see fig. 4.14) and USA (from 76% 1990 to 74% 2022, see ibid.). For the share on total methane emissions, both curves increase over time quite linear, but is the increase for the US mild from 20% (1990) to 25% (2024), is it for DEU quit intense from 1990 also a value of around 20-25% to 2024 with 45%.

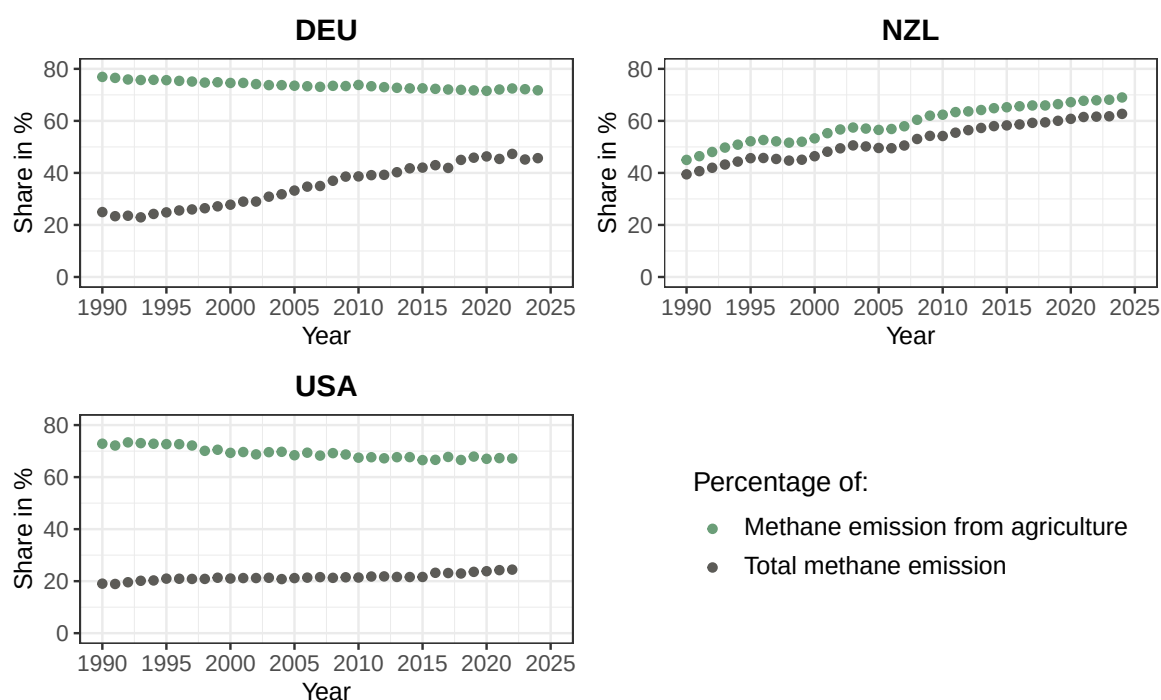


Figure 4.14: Time series of the share of methane emissions resulting from enteric fermentation of cattle on methane produced in the entire agriculture sector and the total amount of CH₄ per country and year. The calculation of the share are based on the information of the reported CRT.

New Zealand time series for the development of the share of enteric fermentation cattle CH₄ emission looks a bit different to those described (see fig. 4.14). Both curves for the share on total and agriculture methane emission show a similar form, they just have slightly different values. The share on agriculture on the one hand grew from 1990 to 2024 from 45% to 70%,

on the other hand the share for total emission from 40% to a bit less than 65%.

4.2.3 Reduction potential in enteric fermentation processes of cattle

Hier muss definitiv in der Diskussion ein Vergleich mit der atmospheric growth rate von 20.1-21.7 Tg CH₄ / year für diese dekade kommen -> in sangios et al/ siehe präsi + kann hier auch darüber sprechen, dass linearität von Ym eindeutig den von DE überwiegt

Figure 4.15 shows the average reduction in CH₄ emissions during the decade from 2010 to 2019 in the countries NZL, DEU, and USA for various possible scenarios in which the shift in the rumen microbiome was successful to varying degrees. The scenarios are 10, 20, 30, 40, and 50, where the numbers refer to the percentage by which the microbiome is shifted toward non-methane-producing microbes. Table B10 in the appendix lists the precisely calculated amounts of saved CO₂ eq. and CH₄.

For the various scenarios, the CH₄ diagram shows that, under identical scenarios, by far the largest amount of CH₄ can be saved in the United States, and that Germany and New Zealand make roughly equal contributions over the ten-year period examined, with Germany contributing slightly more. Overall, a 10% shift would result in annual emissions of approximately 0.9 to 1 Tg CH₄ less over the decade under consideration, with about 0.75 Tg attributable to the United States alone.

At 30%, the total reduction in methane emissions would be approximately 2.75, and at 50%, it would be almost exactly 4.5 Tg. Table B10 However, the appendix shows that the reductions are not entirely linear but follow a slightly nonlinear trend, even though this is depicted differently in fig. 4.15 (for example, DEU reaches 104.40 Tg with a 10% shift and 206.52 Tg CH₄ with a 20% shift).

For global warming potential (GWP), significant quantitative differences can be observed depending on the amount of gas emitted. For example, in a “scenario with a 30% emission reduction” over a 20-year time horizon, approximately 220 Tg less eq. CO₂ would be emitted across all countries combined; over a 100-year time horizon, this figure would be only 75 Tg. The relative dominance of the United States remains evident in these figures.

4.2.4 Prediction for 2030

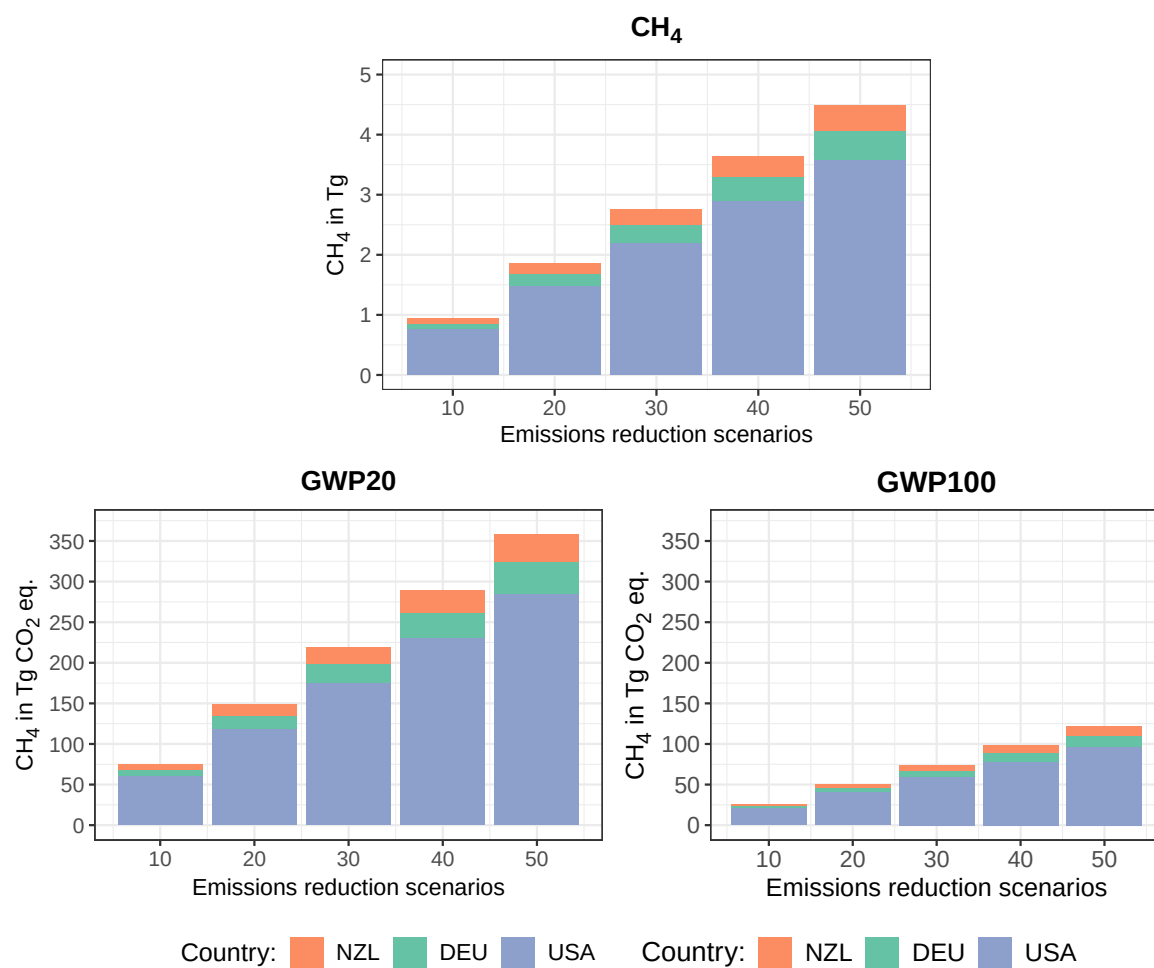


Figure 4.15: Saved CH₄ through different scenarios (10, 20, 30, 40 and 50%) of a successful shift of rumen microbiome towards less methanogenic microbes. The saved methane is plotted as Tg CH₄ against the these five "Emission reduction scenarios" (CH₄). The values for the global warming potential (GWP) for 20 and 100 years are plotted as well. The saved CH₄ is calculated as an average for the decade 2010-2019 for each country (Germany (DEU), New Zealand (NZL) and United States of America (USA))

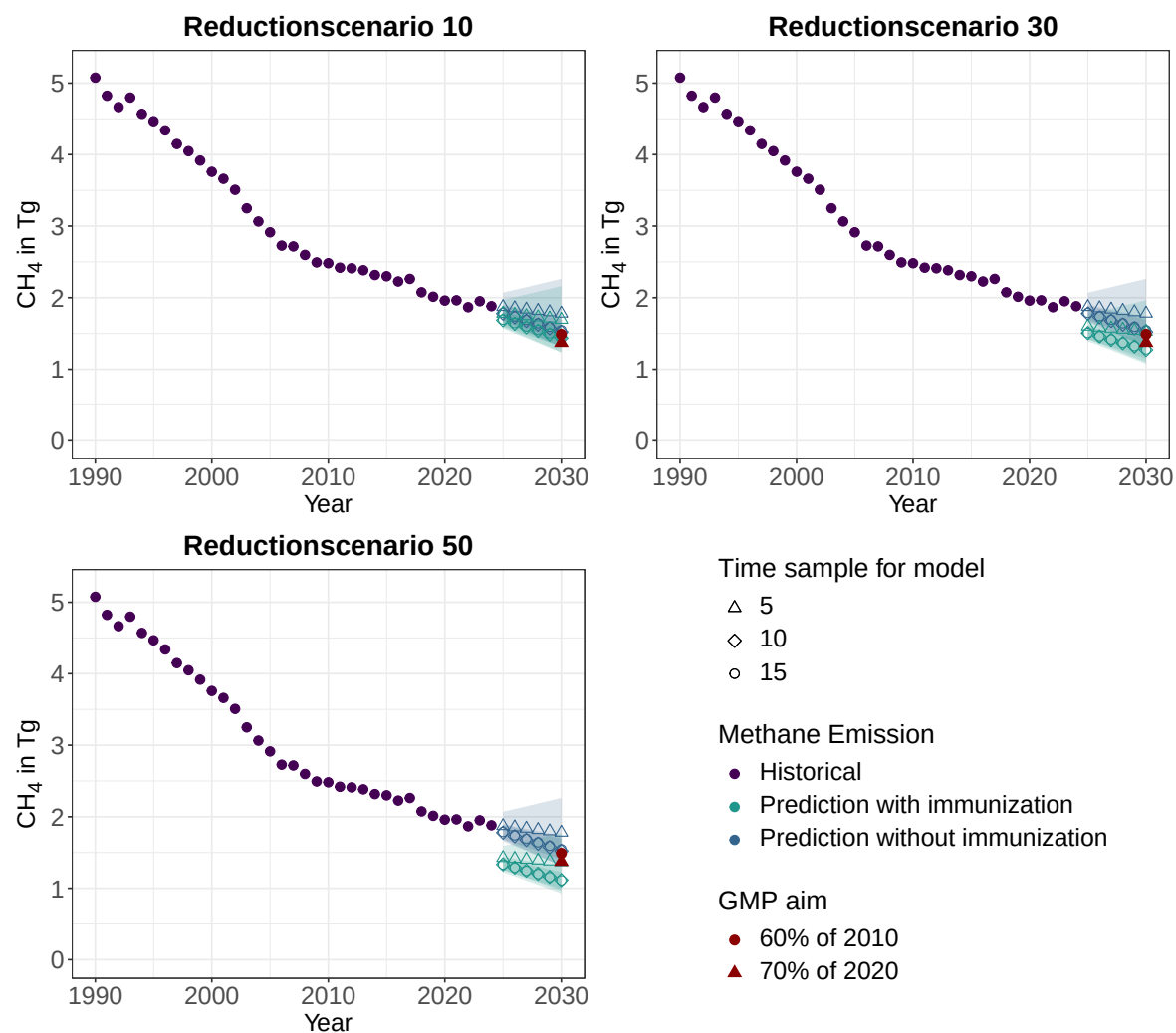


Figure 4.16: DEU

5 DISCUSSION

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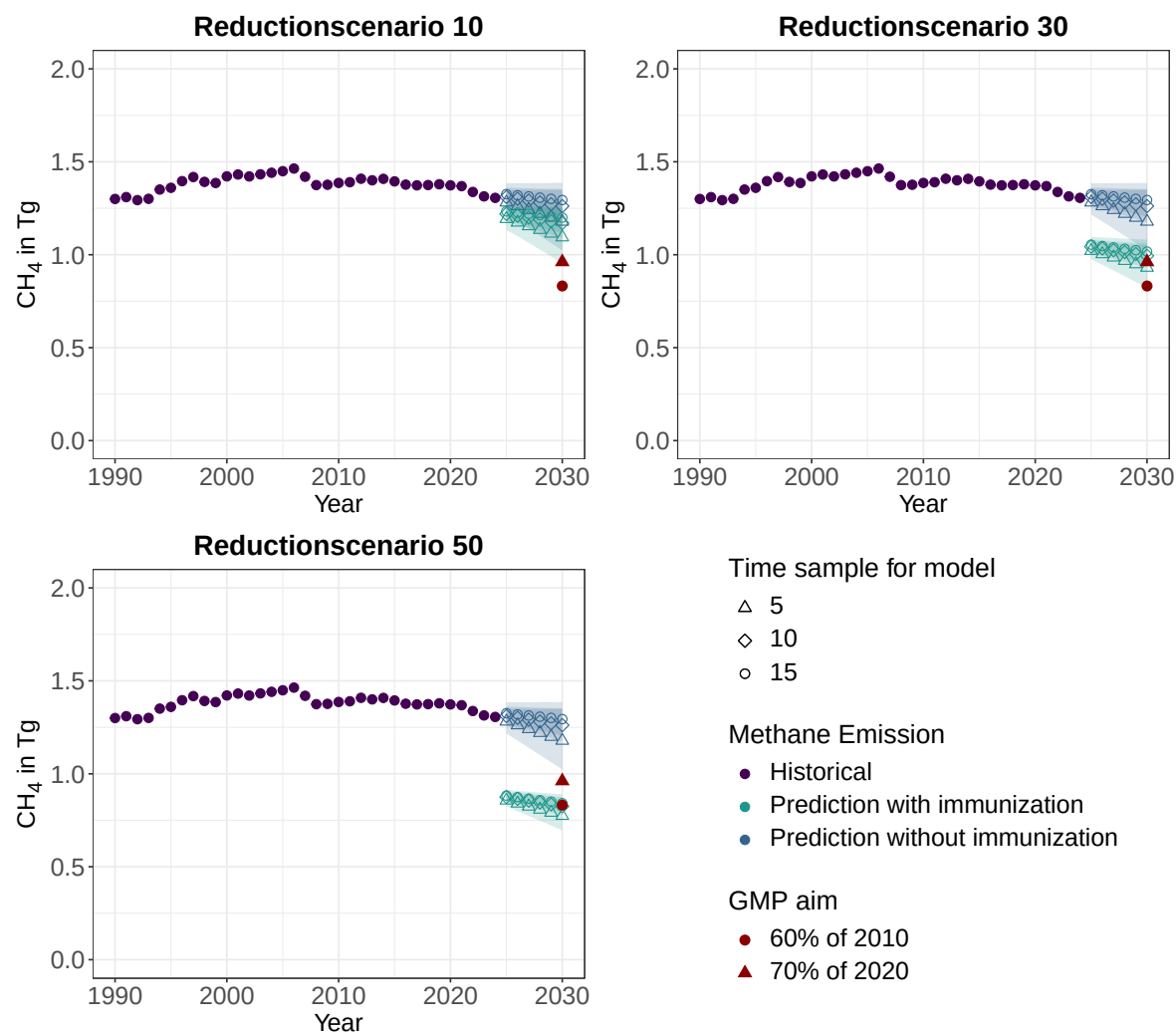


Figure 4.17: NZL

justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris. Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent eismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

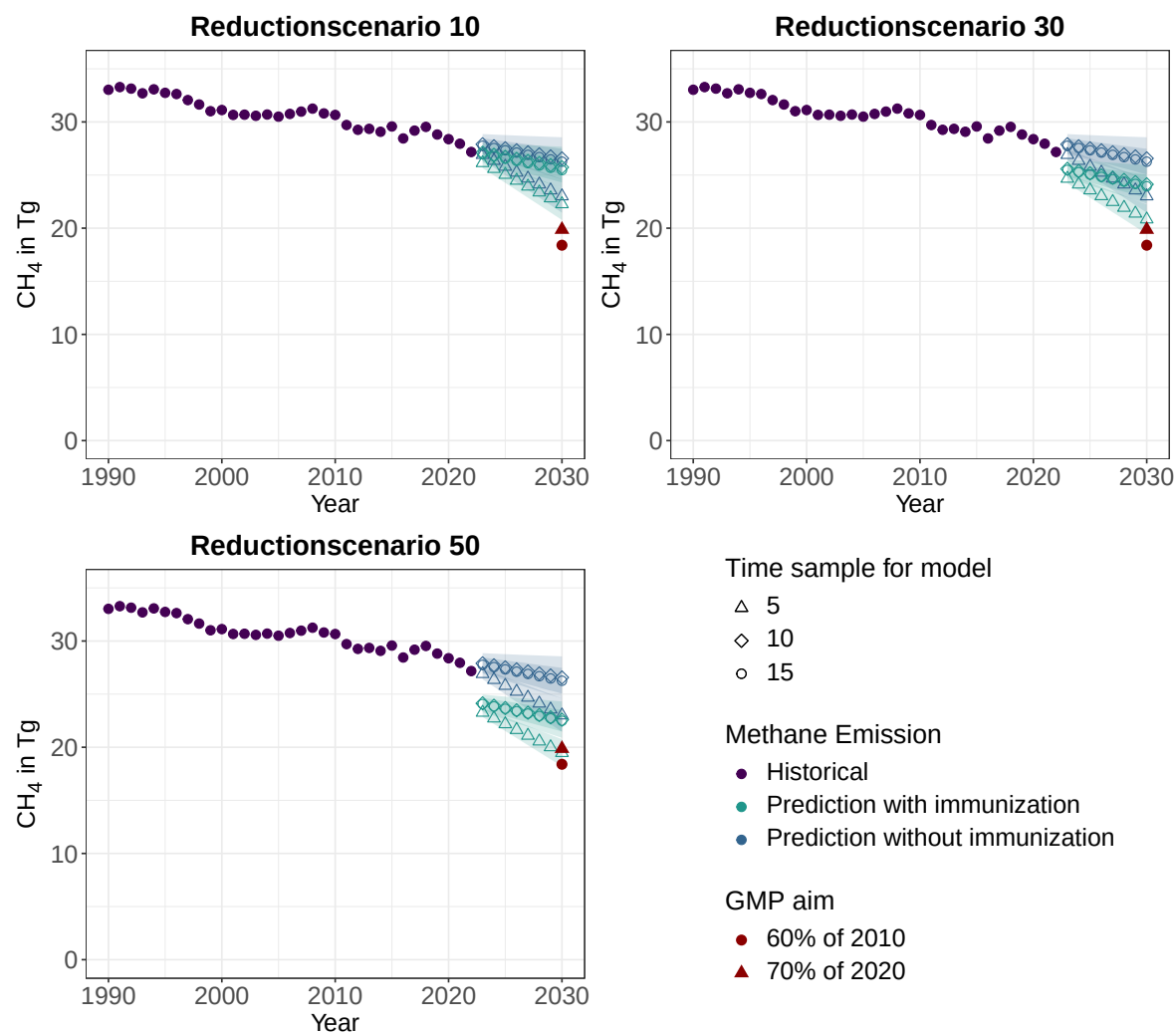


Figure 4.18: USA

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc.

Nam feugiat lacus vel est. Curabitur consectetur.

6 CONCLUSION

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

BIOCHEMICAL TERMS

β -sheet	A common secondary structure of proteins. It consists of β -strands arranged in parallel or antiparallel orientations. β -strands are illustrated as arrows. Another typical secondary structure is the α -helix.
16S-Sequencing	16S sequencing targets the 16s genes, which code for the 16S ribosomal RNA (rRNA). The 16S rRNA is a part of the prokaryotic ribosome. These 16-s base sequence is specific for the organism.
AlphaFold	Is a KI based software for prediction of 3-D structure of proteins. For the prediction it needs an amino acid sequence.
archaea	Archaea constitute one of the three major domains of life (bacteria, archaea, eukaryotes), and, like bacteria, they belong to the prokaryotes (Fritzsche, 2024 , p. 3).
C-terminal end	The end of the amino acid sequence of a protein, where the carboxy group of the terminal amino acid has no covalent connection to another amino acid.
colony	In a microbiological context, a colony is defined as a clearly distinct cluster of cells that can be traced back to a single parent cell through cell division.
complex media	Complex media contain natural, organic molecules, that are not precisely defined chemically (e.g. yeast extract or pepton) So a lot of different microbes can grow on them (Fritzsche, 2024 , p. 359).

glovebox	A container that is gas-tight, so that a different atmosphere can be created.
ORI	Origin of replication, is the region of the DNA, where the machinery responsible for replicating the entire genomic DNA first binds and begins replication.
phase-contrast microscope	Phase-contrast microscopy is based on the fact, that objects or medium not only changes the amplitude of light but also the phase of it. The phase-contrast microscope makes this phenomena visual for the human eye (Pawlizak, 2009).
plasmid	A small ring of double-stranded DNA. Occurs naturally in archaea and bacteria, but is also used for scientific purposes. Here often as a vector for a transformation.
Plating	Cultivate microorganism on a solid medium (contains a gelling agent like agar). The way is the plating the process how the organism get distributed on the plate / solid media. Different technics are possible like spreading or pouring (Fritsche, 2024 , p. 364).
pour plating	A plating technique in which the solution containing microorganisms is mixed with the agar-containing culture medium while it is still liquid (at about 45 °C). This mixture is then poured into a Petri dish, where it solidifies(Fritsche, 2024 , p. 364).

promotor	Region on DNA, which allows binding of relevant enzymes for the transcription of genes. So it is a essential region for gene expression.
restricitionsite	Restrictionsites are a specific sequence of approx. 4 to 10 bp that are recognized by a specific restriction-enzyme. This cuts the double-stranded DNA at this site in a specific behavior.
spread plating	A plating technique in which the solution containing microorganisms is put on the solid agar-containing culture medium. The solution is then spread across the agar plate using a Dragalski spatula (Fritsche, 2024 , p. 364).
transformation	The process of getting a extern DNA into the target organism and its expression. If the target organism is a animal eucaryotic cell it is called transfection.

GEOGRAPHICAL TERMS

CRT Common reporting table is a template for countries to report its emission of GHG fitting IPCC standards. The UNCFF has obliged the ANNEX-I countries to publish this every year since 1990.

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Kleinen, T., Krummel, P. B., Lauerwald, R., Li, T., Liu, X., McDonald, K. C., Melton, J. R., Mühle, J., Müller, J., Murguía-Flores, F., Niwa, Y., Noce, S., Pan, S., Parker, R. J., Peng, C., Ramonet, M., Riley, W. J., Rocher-Ros, G., Rosentreter, J. A., Sasakawa, M., Segers, A., Smith, S. J., Stanley, E. H., Thanwerdas, J., Tian, H., Tsuruta, A., Tubiello, F. N., Weber, T. S., van der Werf, G. R., Worthy, D. E. J., Xi, Y., Yoshida, Y., Zhang, W., Zheng, B., Zhu, Q., Zhu, Q., and Zhuang, Q. (2025). Global Methane Budget 2000–2020. *Earth System Science Data*, 17(5):1873–1958.

APPENDIX

A Figures

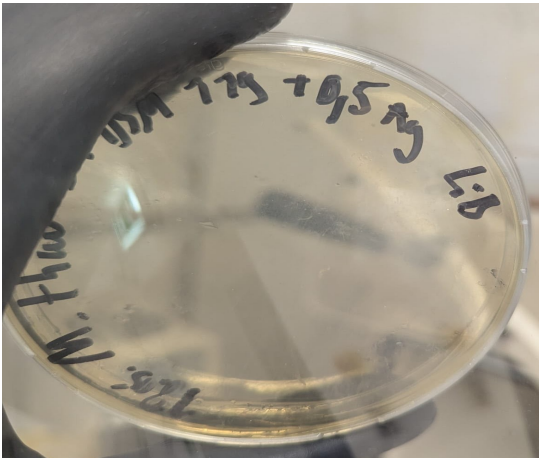


Figure A1: Pour plating result of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) DSM 119 (1% agar). 100 μ l of a full-grown culture have been plated. Cultivated in a jar with H_2/CO_2 gas phase. No growth detected.

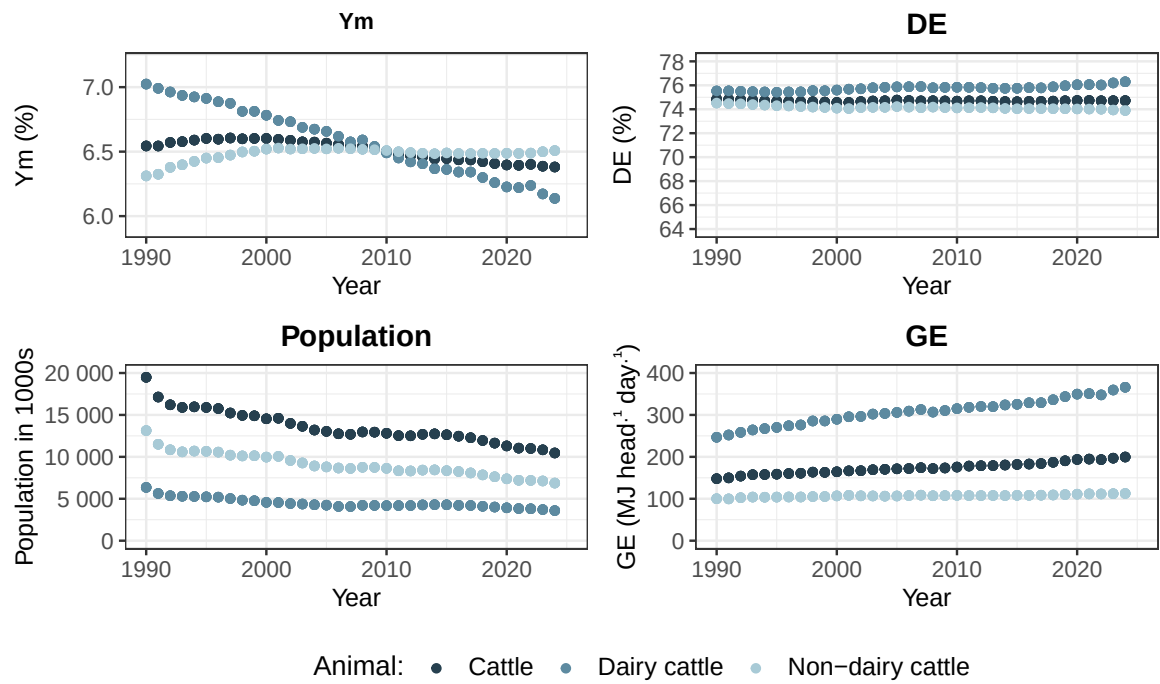


Figure A2: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for Germany. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

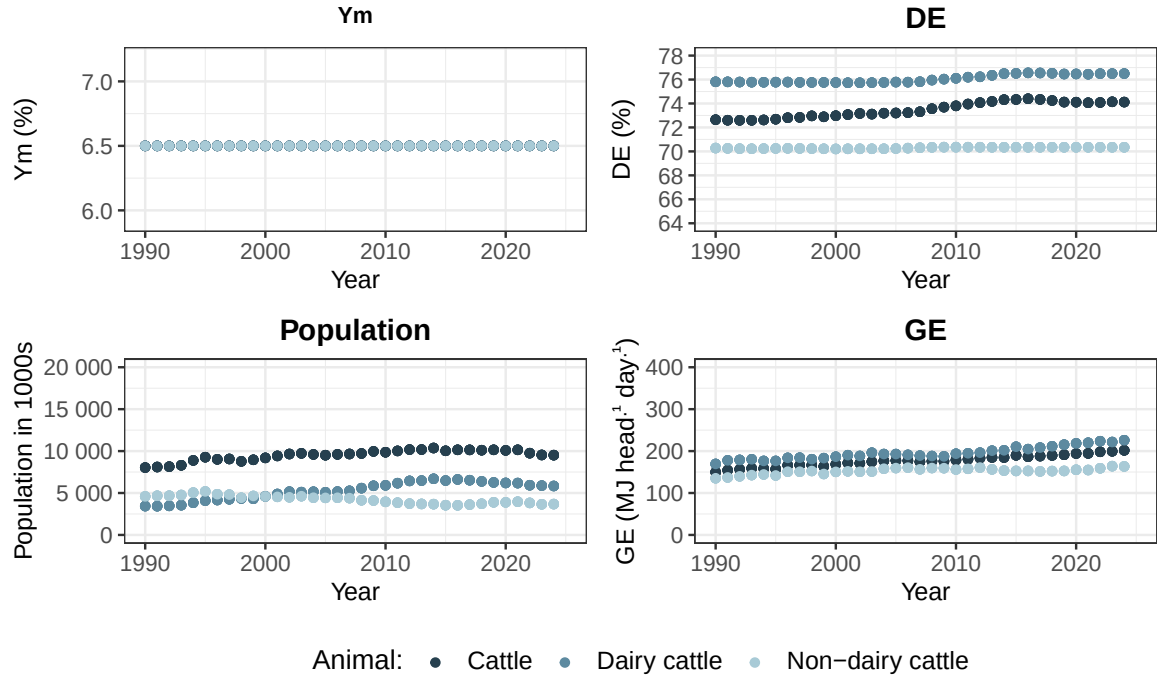


Figure A3: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for New Zealand. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

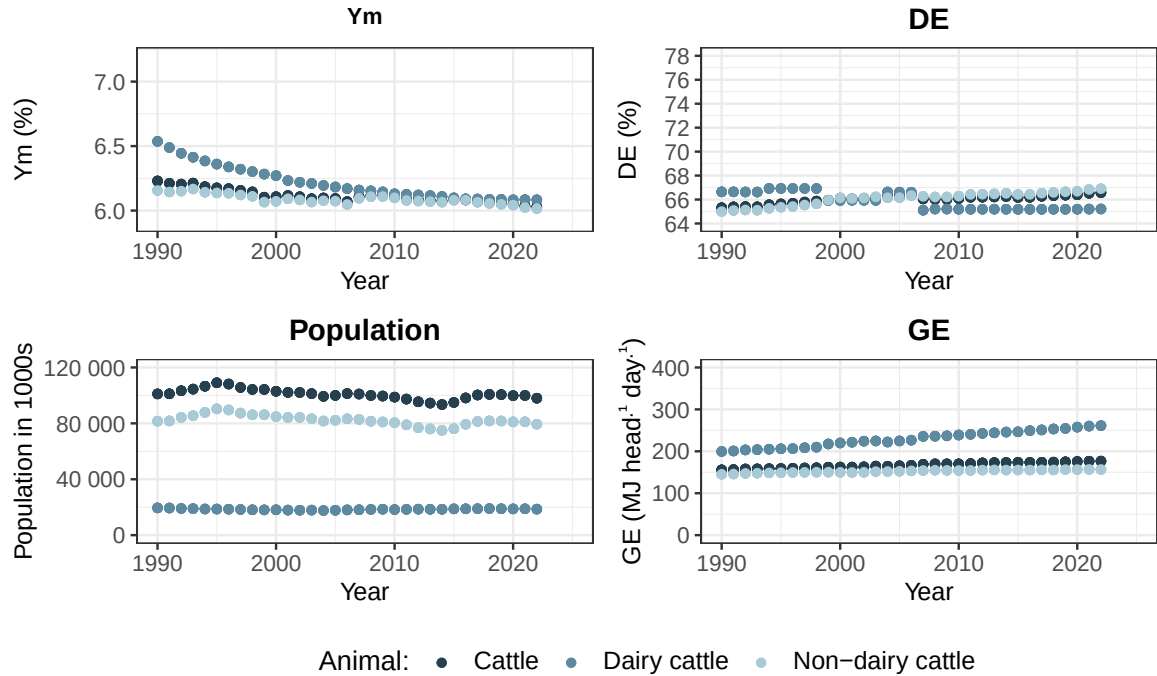


Figure A4: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for United States of America. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

B Tables

Table B1: Counted cells for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) with an OD (600 nm) of 0.387. Cells were counted with a phase-contrast microscope and a Buerk counting chamber at 400x magnification. Per chamber filling four square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	138	159	226	233	189
2	260	271	293	265	272.25
3	180	197	216	188	195.25
4	207	191	211	227	209

Table B2: Counted cells for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) with an OD (600 nm) of 0.396. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	177	239	263	254	233.25
2	246	241	267	208	240.5
3	175	176	161	221	183.25
4	217	185	205	180	196.75

Table B3: Counted cells for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) with an OD (600 nm) of 0.235. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	189	232	250	265	234
2	237	209	224	263	233.25
3	186	253	230	237	226.5
4	146	187	216	262	202.75

Table B8: bovi

culture	day	conc. [μ g/ml]	OD600	OD600 adj.
Cltr. 1	0	0	-0.002	0.001
Cltr. 1	0	1	-0.002	0.001
Cltr. 1	0	2	-0.003	0.001

Cltr. 1	0	5	0.016	0.016
Cltr. 1	0	10	0.008	0.008
Cltr. 1	0	20	0.004	0.004
Cltr. 1	1	0	0.021	0.021
Cltr. 1	1	1	0.026	0.026
Cltr. 1	1	2	0.021	0.021
Cltr. 1	1	5	0.030	0.030
Cltr. 1	1	10	0.004	0.004
Cltr. 1	1	20	-0.008	0.001
Cltr. 1	2	0	0.131	0.131
Cltr. 1	2	1	0.187	0.187
Cltr. 1	2	2	0.149	0.149
Cltr. 1	2	5	0.073	0.073
Cltr. 1	2	10	-0.012	0.001
Cltr. 1	2	20	-0.020	0.001
Cltr. 1	4	0	0.212	0.212
Cltr. 1	4	1	0.315	0.315
Cltr. 1	4	2	0.194	0.194
Cltr. 1	4	5	0.155	0.155
Cltr. 1	4	10	-0.014	0.001
Cltr. 1	4	20	-0.018	0.001
Cltr. 1	7	0	0.176	0.176
Cltr. 1	7	1	0.226	0.226
Cltr. 1	7	2	0.140	0.140
Cltr. 1	7	5	0.166	0.166
Cltr. 1	7	10	0.189	0.189
Cltr. 1	7	20	-0.012	0.001
Cltr. 1	8	0	0.163	0.163
Cltr. 1	8	1	0.207	0.207
Cltr. 1	8	2	0.132	0.132
Cltr. 1	8	5	0.224	0.224
Cltr. 1	8	10	0.181	0.181
Cltr. 1	8	20	-0.004	0.001
Cltr. 1	9	0	0.148	0.148

Cltr. 1	9	1	0.196	0.196
Cltr. 1	9	2	0.117	0.117
Cltr. 1	9	5	0.193	0.193
Cltr. 1	9	10	0.170	0.170
Cltr. 1	9	20	0.044	0.044
Cltr. 1	10	0	0.111	0.111
Cltr. 1	10	1	0.167	0.167
Cltr. 1	10	2	0.095	0.095
Cltr. 1	10	5	0.163	0.163
Cltr. 1	10	10	0.137	0.137
Cltr. 1	10	20	0.146	0.146
Cltr. 1	11	0	0.082	0.082
Cltr. 1	11	1	0.152	0.152
Cltr. 1	11	2	0.066	0.066
Cltr. 1	11	5	0.134	0.134
Cltr. 1	11	10	0.109	0.109
Cltr. 1	11	20	0.111	0.111
Cltr. 2	0	0	0.006	0.006
Cltr. 2	0	1	0.008	0.008
Cltr. 2	0	2	0.002	0.002
Cltr. 2	0	5	0.027	0.027
Cltr. 2	0	10	0.003	0.003
Cltr. 2	0	20	0.002	0.002
Cltr. 2	1	0	0.057	0.057
Cltr. 2	1	1	0.060	0.060
Cltr. 2	1	2	0.049	0.049
Cltr. 2	1	5	0.037	0.037
Cltr. 2	1	10	-0.002	0.001
Cltr. 2	1	20	-0.009	0.001
Cltr. 2	2	0	0.189	0.189
Cltr. 2	2	1	0.490	0.490
Cltr. 2	2	2	0.245	0.245
Cltr. 2	2	5	0.091	0.091
Cltr. 2	2	10	-0.013	0.001

Cltr. 2	2	20	-0.022	0.001
Cltr. 2	4	0	0.254	0.254
Cltr. 2	4	1	0.400	0.400
Cltr. 2	4	2	0.322	0.322
Cltr. 2	4	5	0.051	0.051
Cltr. 2	4	10	-0.020	0.001
Cltr. 2	4	20	-0.024	0.001
Cltr. 2	7	0	0.193	0.193
Cltr. 2	7	1	0.270	0.270
Cltr. 2	7	2	0.217	0.217
Cltr. 2	7	5	0.069	0.069
Cltr. 2	7	10	0.007	0.007
Cltr. 2	7	20	-0.016	0.001
Cltr. 2	8	0	0.187	0.187
Cltr. 2	8	1	0.248	0.248
Cltr. 2	8	2	0.200	0.200
Cltr. 2	8	5	0.130	0.130
Cltr. 2	8	10	0.128	0.128
Cltr. 2	8	20	-0.014	0.001
Cltr. 2	9	0	0.157	0.157
Cltr. 2	9	1	0.233	0.233
Cltr. 2	9	2	0.190	0.190
Cltr. 2	9	5	0.113	0.113
Cltr. 2	9	10	0.187	0.187
Cltr. 2	9	20	-0.017	0.001
Cltr. 2	10	0	0.123	0.123
Cltr. 2	10	1	0.204	0.204
Cltr. 2	10	2	0.163	0.163
Cltr. 2	10	5	0.091	0.091
Cltr. 2	10	10	0.141	0.141
Cltr. 2	10	20	-0.023	0.001
Cltr. 2	11	0	0.095	0.095
Cltr. 2	11	1	0.176	0.176
Cltr. 2	11	2	0.148	0.148

Cltr. 2	11	5	0.079	0.079
Cltr. 2	11	10	0.117	0.117
Cltr. 2	11	20	-0.002	0.001
Cltr. 3	0	0	0.000	0.001
Cltr. 3	0	1	-0.004	0.001
Cltr. 3	0	2	-0.004	0.001
Cltr. 3	0	5	-0.004	0.001
Cltr. 3	0	10	0.007	0.007
Cltr. 3	0	20	0.008	0.008
Cltr. 3	1	0	0.006	0.006
Cltr. 3	1	1	-0.002	0.001
Cltr. 3	1	2	0.002	0.002
Cltr. 3	1	5	-0.008	0.001
Cltr. 3	1	10	-0.010	0.001
Cltr. 3	1	20	-0.004	0.001
Cltr. 3	2	0	0.183	0.183
Cltr. 3	2	1	0.146	0.146
Cltr. 3	2	2	0.450	0.450
Cltr. 3	2	5	0.010	0.010
Cltr. 3	2	10	-0.017	0.001
Cltr. 3	2	20	-0.013	0.001
Cltr. 3	4	0	0.254	0.254
Cltr. 3	4	1	0.300	0.300
Cltr. 3	4	2	0.390	0.390
Cltr. 3	4	5	0.002	0.002
Cltr. 3	4	10	-0.020	0.001
Cltr. 3	4	20	-0.009	0.001
Cltr. 3	7	0	0.206	0.206
Cltr. 3	7	1	0.234	0.234
Cltr. 3	7	2	0.248	0.248
Cltr. 3	7	5	0.167	0.167
Cltr. 3	7	10	-0.006	0.001
Cltr. 3	7	20	-0.002	0.001
Cltr. 3	8	0	0.174	0.174

Cltr. 3	8	1	0.219	0.219
Cltr. 3	8	2	0.213	0.213
Cltr. 3	8	5	0.168	0.168
Cltr. 3	8	10	0.009	0.009
Cltr. 3	8	20	0.002	0.002
Cltr. 3	9	0	0.161	0.161
Cltr. 3	9	1	0.202	0.202
Cltr. 3	9	2	0.203	0.203
Cltr. 3	9	5	0.146	0.146
Cltr. 3	9	10	0.073	0.073
Cltr. 3	9	20	0.004	0.004
Cltr. 3	10	0	0.120	0.120
Cltr. 3	10	1	0.165	0.165
Cltr. 3	10	2	0.175	0.175
Cltr. 3	10	5	0.115	0.115
Cltr. 3	10	10	0.100	0.100
Cltr. 3	10	20	0.116	0.116
Cltr. 3	11	0	0.096	0.096
Cltr. 3	11	1	0.142	0.142
Cltr. 3	11	2	0.159	0.159
Cltr. 3	11	5	0.091	0.091
Cltr. 3	11	10	0.072	0.072
Cltr. 3	11	20	0.130	0.130

Table B9: thau

culture	day	conc. [$\mu\text{g/ml}$]	OD600	OD600 adj.
Cltr. 1	0	0	0.006	0.006
Cltr. 1	0	1	0.007	0.007
Cltr. 1	0	2	0.003	0.003
Cltr. 1	0	5	0.004	0.004
Cltr. 1	0	10	0.004	0.004
Cltr. 1	0	20	0.002	0.002

Cltr. 1	1	0	0.017	0.017
Cltr. 1	1	1	0.036	0.036
Cltr. 1	1	2	0.019	0.019
Cltr. 1	1	5	-0.008	0.001
Cltr. 1	1	10	-0.017	0.001
Cltr. 1	1	20	-0.023	0.001
Cltr. 1	3	0	0.103	0.103
Cltr. 1	3	1	0.180	0.180
Cltr. 1	3	2	0.125	0.125
Cltr. 1	3	5	-0.002	0.001
Cltr. 1	3	10	-0.019	0.001
Cltr. 1	3	20	-0.023	0.001
Cltr. 1	6	0	0.055	0.055
Cltr. 1	6	1	0.078	0.078
Cltr. 1	6	2	0.110	0.110
Cltr. 1	6	5	-0.001	0.001
Cltr. 1	6	10	-0.015	0.001
Cltr. 1	6	20	-0.019	0.001
Cltr. 1	7	0	0.030	0.030
Cltr. 1	7	1	0.054	0.054
Cltr. 1	7	2	0.091	0.091
Cltr. 1	7	5	-0.003	0.001
Cltr. 1	7	10	-0.019	0.001
Cltr. 1	7	20	-0.019	0.001
Cltr. 1	8	0	0.018	0.018
Cltr. 1	8	1	0.035	0.035
Cltr. 1	8	2	0.076	0.076
Cltr. 1	8	5	-0.009	0.001
Cltr. 1	8	10	-0.021	0.001
Cltr. 1	8	20	-0.022	0.001
Cltr. 1	9	0	-0.002	0.001
Cltr. 1	9	1	0.008	0.008
Cltr. 1	9	2	0.043	0.043
Cltr. 1	9	5	-0.024	0.001

Cltr. 1	9	10	-0.035	0.001
Cltr. 1	9	20	-0.035	0.001
Cltr. 1	10	0	-0.013	0.001
Cltr. 1	10	1	-0.009	0.001
Cltr. 1	10	2	0.027	0.027
Cltr. 1	10	5	-0.029	0.001
Cltr. 1	10	10	-0.037	0.001
Cltr. 1	10	20	-0.038	0.001
Cltr. 2	0	0	0.003	0.003
Cltr. 2	0	1	0.001	0.001
Cltr. 2	0	2	0.004	0.004
Cltr. 2	0	5	0.000	0.001
Cltr. 2	0	10	0.002	0.002
Cltr. 2	0	20	0.014	0.014
Cltr. 2	1	0	0.041	0.041
Cltr. 2	1	1	0.036	0.036
Cltr. 2	1	2	-0.004	0.001
Cltr. 2	1	5	-0.005	0.001
Cltr. 2	1	10	-0.013	0.001
Cltr. 2	1	20	-0.003	0.001
Cltr. 2	3	0	0.191	0.191
Cltr. 2	3	1	0.260	0.260
Cltr. 2	3	2	0.029	0.029
Cltr. 2	3	5	0.001	0.001
Cltr. 2	3	10	-0.011	0.001
Cltr. 2	3	20	-0.007	0.001
Cltr. 2	6	0	0.089	0.089
Cltr. 2	6	1	0.150	0.150
Cltr. 2	6	2	0.016	0.016
Cltr. 2	6	5	0.004	0.004
Cltr. 2	6	10	-0.010	0.001
Cltr. 2	6	20	0.000	0.001
Cltr. 2	7	0	0.056	0.056
Cltr. 2	7	1	0.110	0.110

Cltr. 2	7	2	0.017	0.017
Cltr. 2	7	5	0.000	0.001
Cltr. 2	7	10	-0.015	0.001
Cltr. 2	7	20	-0.003	0.001
Cltr. 2	8	0	0.033	0.033
Cltr. 2	8	1	0.078	0.078
Cltr. 2	8	2	0.007	0.007
Cltr. 2	8	5	-0.007	0.001
Cltr. 2	8	10	-0.015	0.001
Cltr. 2	8	20	-0.007	0.001
Cltr. 2	9	0	0.003	0.003
Cltr. 2	9	1	0.036	0.036
Cltr. 2	9	2	0.004	0.004
Cltr. 2	9	5	-0.021	0.001
Cltr. 2	9	10	-0.026	0.001
Cltr. 2	9	20	-0.019	0.001
Cltr. 2	10	0	-0.010	0.001
Cltr. 2	10	1	0.008	0.008
Cltr. 2	10	2	-0.012	0.001
Cltr. 2	10	5	-0.027	0.001
Cltr. 2	10	10	-0.038	0.001
Cltr. 2	10	20	-0.024	0.001
Cltr. 3	0	0	0.016	0.016
Cltr. 3	0	1	0.003	0.003
Cltr. 3	0	2	0.006	0.006
Cltr. 3	0	5	0.008	0.008
Cltr. 3	0	10	0.022	0.022
Cltr. 3	0	20	0.010	0.010
Cltr. 3	1	0	0.034	0.034
Cltr. 3	1	1	0.002	0.002
Cltr. 3	1	2	-0.002	0.001
Cltr. 3	1	5	-0.020	0.001
Cltr. 3	1	10	-0.008	0.001
Cltr. 3	1	20	-0.023	0.001

Cltr. 3	3	0	0.192	0.192
Cltr. 3	3	1	0.263	0.263
Cltr. 3	3	2	0.067	0.067
Cltr. 3	3	5	0.176	0.176
Cltr. 3	3	10	-0.011	0.001
Cltr. 3	3	20	-0.025	0.001
Cltr. 3	6	0	0.102	0.102
Cltr. 3	6	1	0.183	0.183
Cltr. 3	6	2	0.094	0.094
Cltr. 3	6	5	0.003	0.003
Cltr. 3	6	10	0.000	0.001
Cltr. 3	6	20	-0.012	0.001
Cltr. 3	7	0	0.075	0.075
Cltr. 3	7	1	0.147	0.147
Cltr. 3	7	2	0.085	0.085
Cltr. 3	7	5	0.000	0.001
Cltr. 3	7	10	-0.004	0.001
Cltr. 3	7	20	-0.017	0.001
Cltr. 3	8	0	0.055	0.055
Cltr. 3	8	1	0.112	0.112
Cltr. 3	8	2	0.074	0.074
Cltr. 3	8	5	-0.005	0.001
Cltr. 3	8	10	-0.005	0.001
Cltr. 3	8	20	-0.021	0.001
Cltr. 3	9	0	0.026	0.026
Cltr. 3	9	1	0.076	0.076
Cltr. 3	9	2	0.052	0.052
Cltr. 3	9	5	-0.017	0.001
Cltr. 3	9	10	-0.020	0.001
Cltr. 3	9	20	-0.033	0.001
Cltr. 3	10	0	0.012	0.012
Cltr. 3	10	1	0.054	0.054
Cltr. 3	10	2	0.037	0.037
Cltr. 3	10	5	-0.023	0.001

Cltr. 3	10	10	-0.025	0.001
Cltr. 3	10	20	-0.039	0.001

Table B4: Counted cells for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) with an OD (600 nm) of 0.116. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	255	181	189	171	199
2	127	92	131	193	135.75
3	110	136	153	109	127
4	127	93	109	156	121.25

Table B5: Counted cells for *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) with an OD (600 nm) of 0.378. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	Quadrat_5	$\bar{x}$ cellnumber
1	186	175	173	156	220	182
2	118	204				161
3	211	178	232	211	184	203.2
4	128	128	201	204	160	164.2

Table B6: Counted cells for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) with an OD(600 nm) of 0.287. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	123	162	166	170	155.25
2	162	228	205	242	209.25
3	170	128	137	213	162
4	187	174	263	237	215.25

Table B7: The total average of counted cells ($\bar{x}$ cellnumber) for each *Methanobrevibacter* (*M.*) culture with corresponding OD at 600 nm (OD600). Counting was done with a phase-contrast microscope and a Buerk or Neubauer improved counting chamber at 400x magnification. $\bar{x}$ cellnumber was calculated as arithmetic mean of shown cellcounts in tab. B1 through B6. The dilution factor takes into account the dilution of the culture being counted.

culture	OD600	Dilution factor	$\bar{x}$ cellnumber
<i>M. wolinii</i>	0.387	20	216.375
<i>M. wolinii</i>	0.396	20	213.438
<i>M. thaueri</i>	0.235	10	224.125
<i>M. thaueri</i>	0.116	10	145.750
<i>M. millerae</i>	0.378	20	177.600
<i>M. boviskoreani</i>	0.287	20	185.438

DECLARATION

I herewith declare that I have composed the present thesis myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The thesis in the same or similar form has not been submitted to any examination body and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance. Furthermore, I declare that the submitted written (bound) copies of the present thesis and the version submitted on a data carrier are consistent with each other in contents.

Place, date

Signature

Table B10: Lists saved CH₄ for five scenarios of different success in shifting the microbiom from methanogen towards non-methanogen (10, 20, 30, 40 and 50%). It was calculated an average for the decade 2010 to 2019 for the three countries Germany (DEU), New Zealand (NZL), United States of America (USA). The CO₂ eq. was calculated for the GWP20 and GWP100 with factors taken from ?, Table 7.15, Chapter 7 (79.7 for GWP20 and 27 for GWP100)

country	red.-scenario	CH ₄ [kt]	GWP20 [kt CO ₂ eq.]	GWP100 [kt CO ₂ eq.]
DEU	10	104.40	8 320.60	2 818.77
DEU	20	206.52	16 459.82	5 576.10
DEU	30	306.45	24 424.23	8 274.20
DEU	40	404.27	32 220.05	10 915.20
DEU	50	500.04	39 853.21	13 501.09
NZL	10	88.23	7 031.61	2 382.10
NZL	20	174.47	13 905.60	4 710.81
NZL	30	258.82	20 627.83	6 988.10
NZL	40	341.33	27 203.88	9 215.87
NZL	50	422.07	33 639.03	11 395.91
USA	10	749.55	59 738.75	20 237.72
USA	20	1 479.97	117 953.24	39 959.07
USA	30	2 192.06	174 707.16	59 185.61
USA	40	2 886.58	230 060.64	77 937.73
USA	50	3 564.25	284 070.52	96 234.68